Decision tree

classification & regression

Is it raining outside?

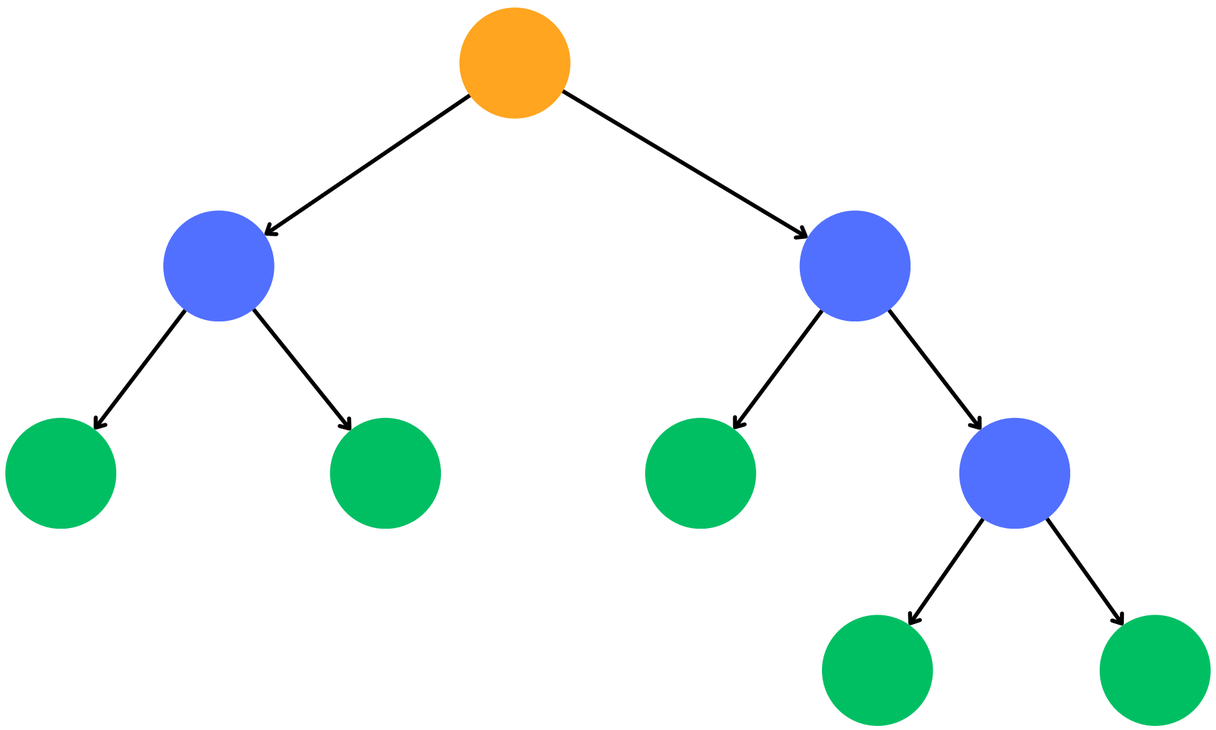
True

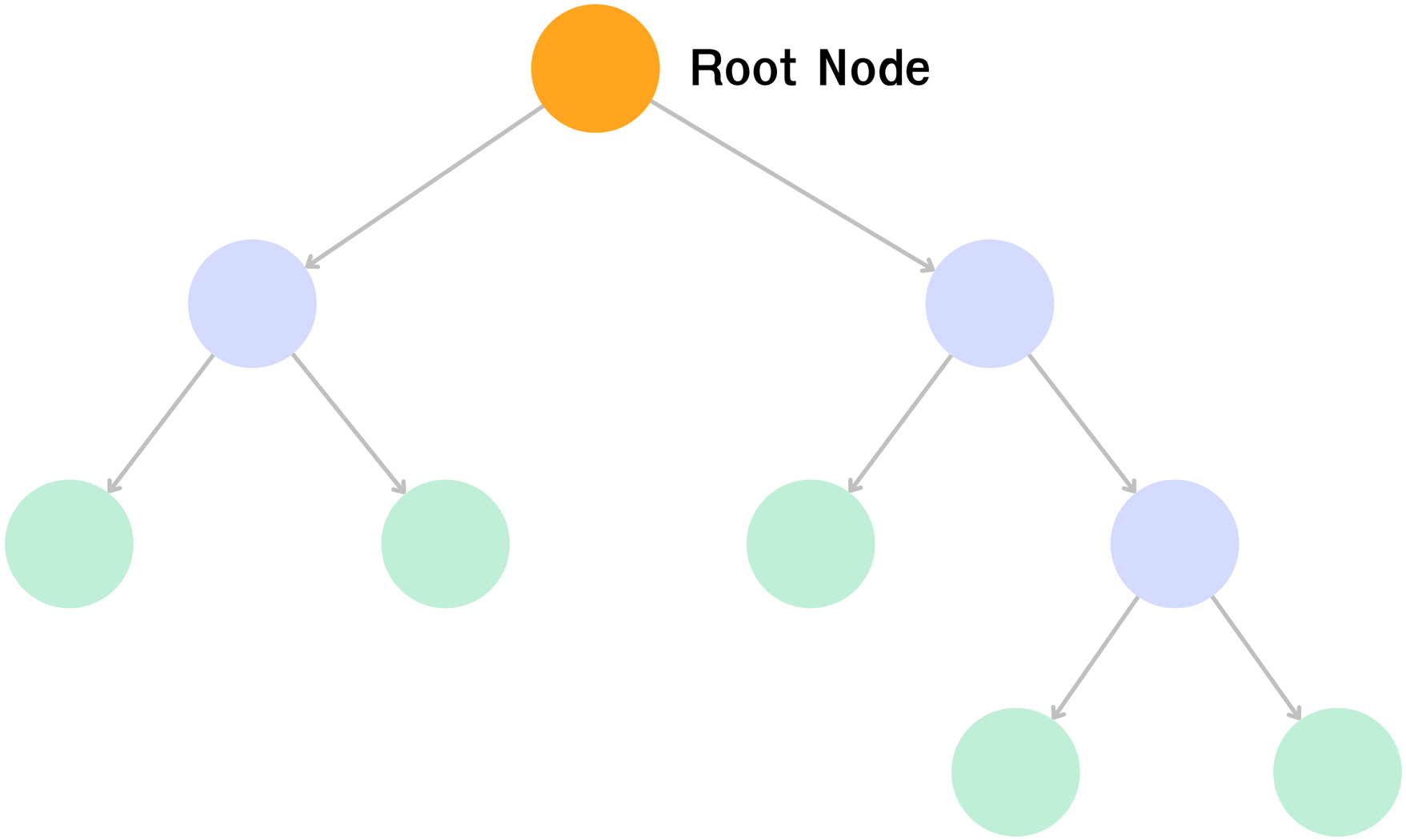
Take an umbrella.

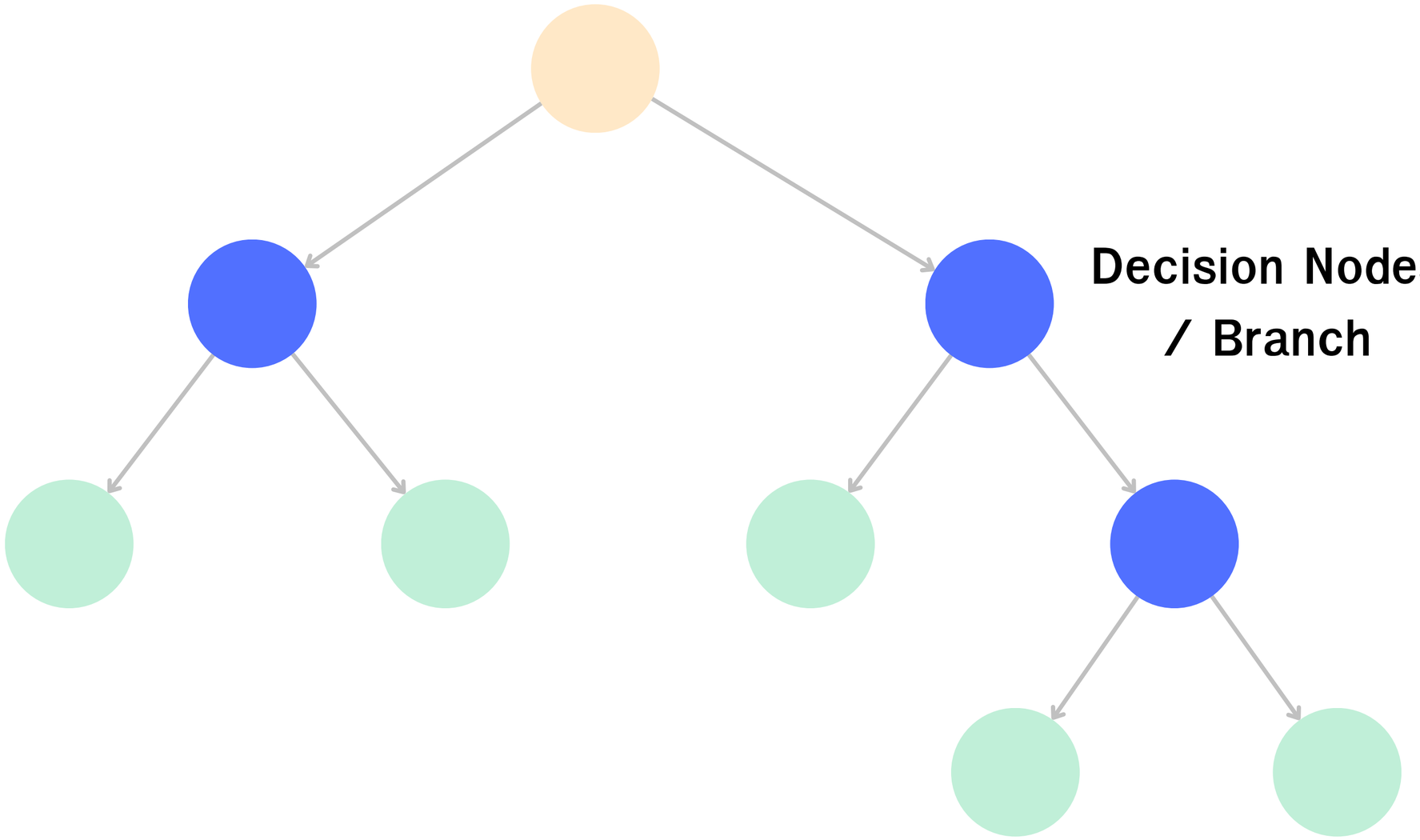
False

Don't take an umbrella.

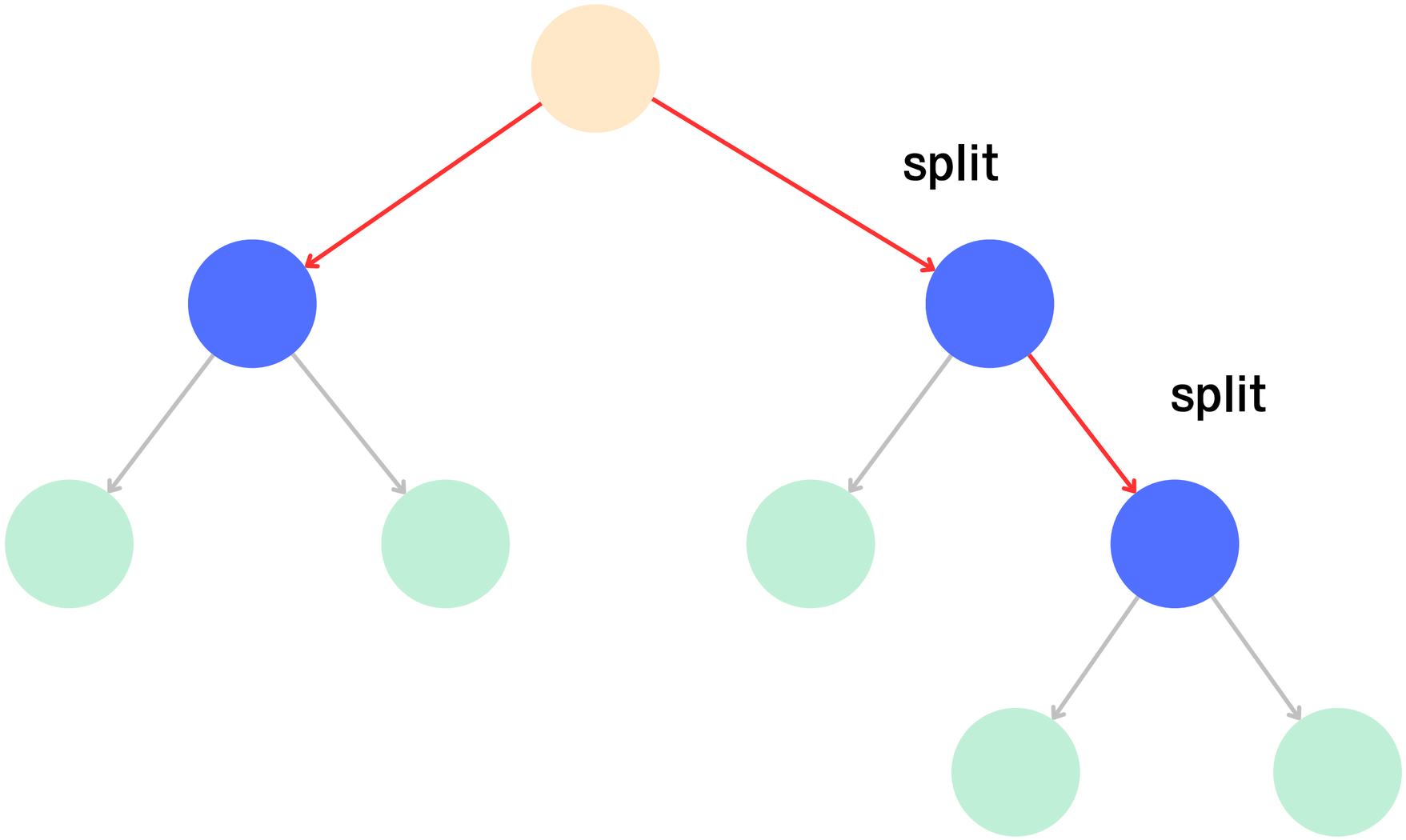
Basic Concepts

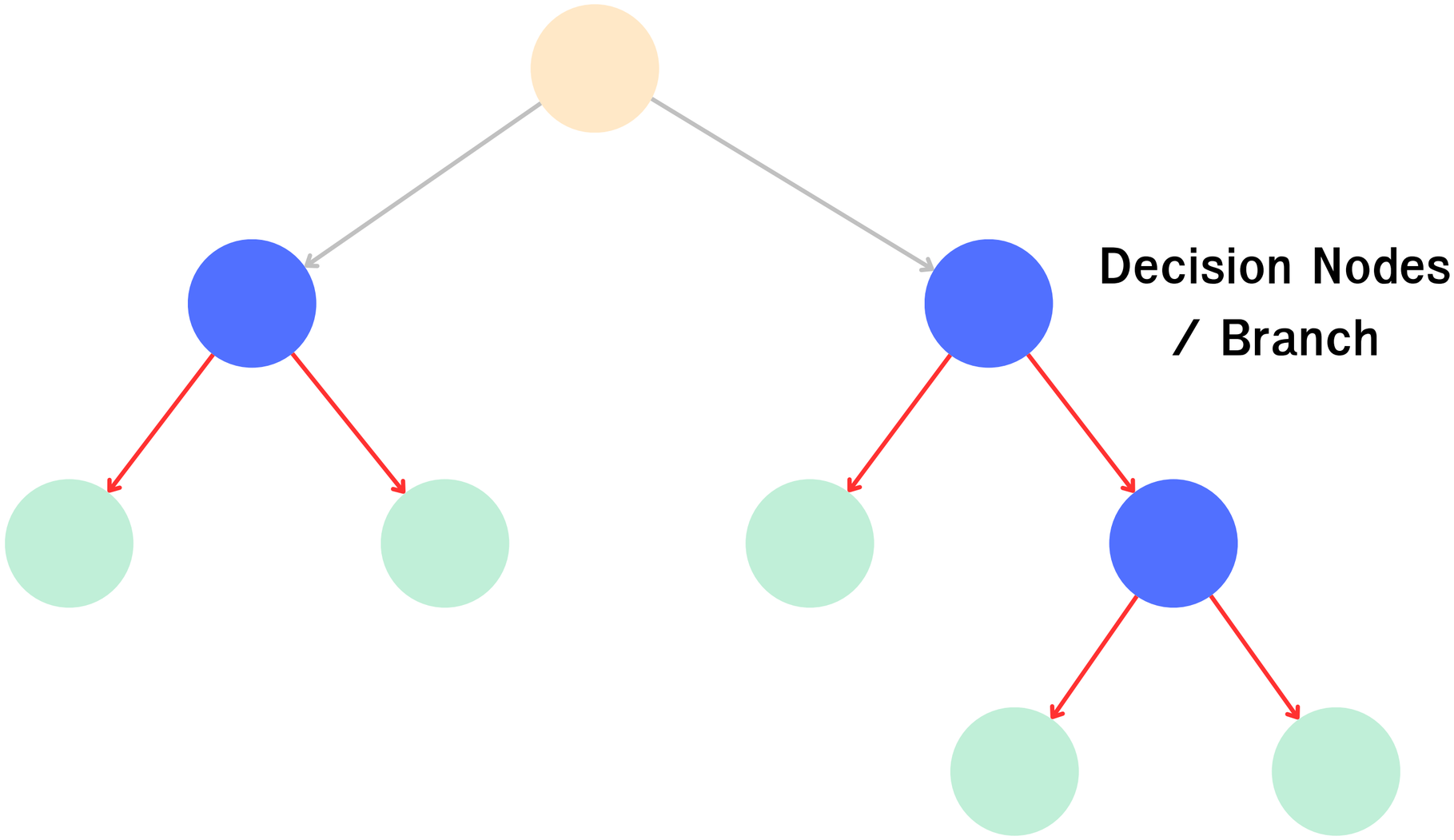


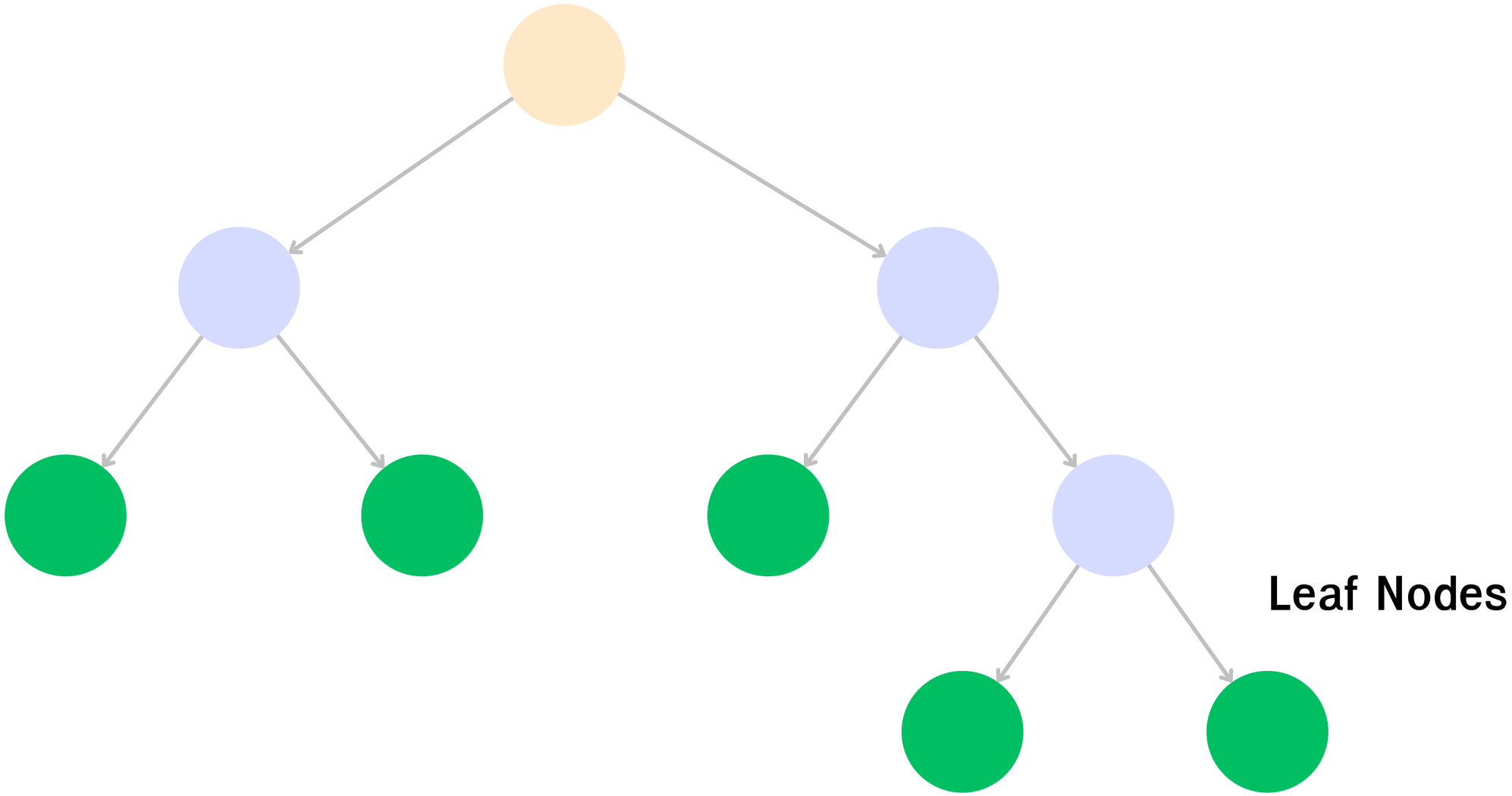

Root Node

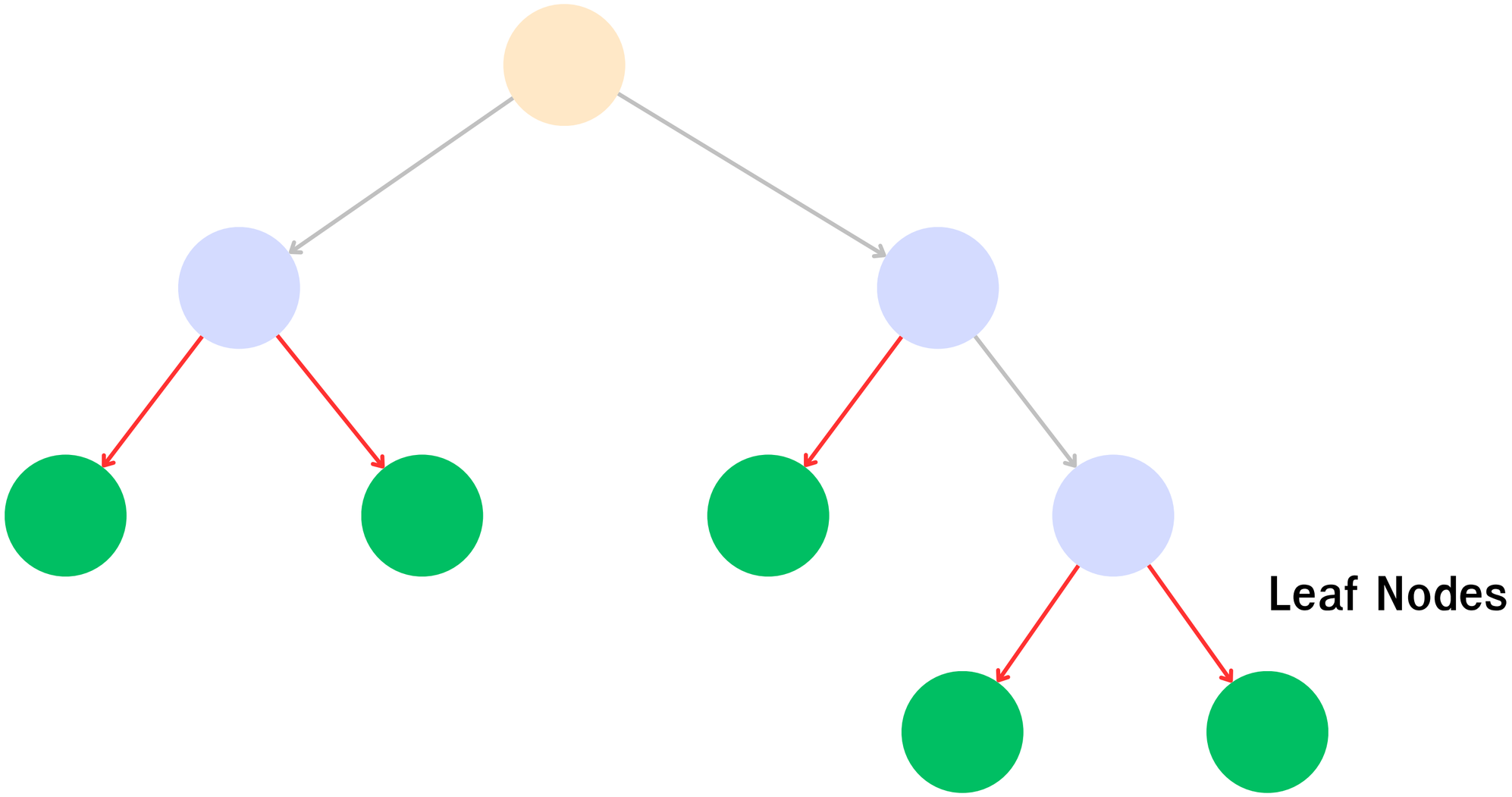


Decision Nodes
/ Branch


split
split


Decision Nodes
/ Branch


Leaf Nodes


Leaf Nodes

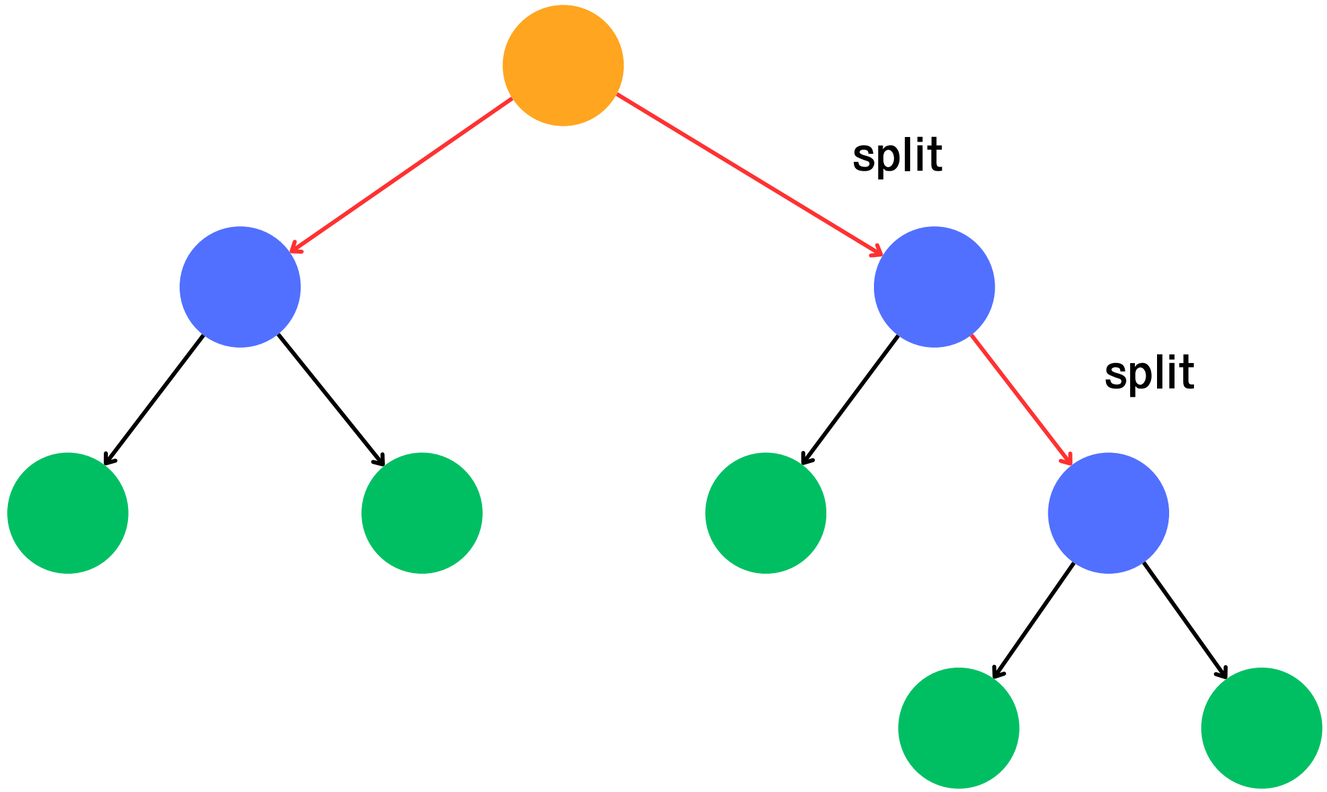
classification

love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

that predicts whether or not someone love Mala.

หลักการสร้างต้นไม้



แบ่งข้อมูล (split) ทีละขั้น ให้ได้ “กิ่ง” ที่ แยกกลุ่มได้ชัดที่สุด (คือข้อมูลในแต่ละกลุ่มมี class เดียวกันมากที่สุด)

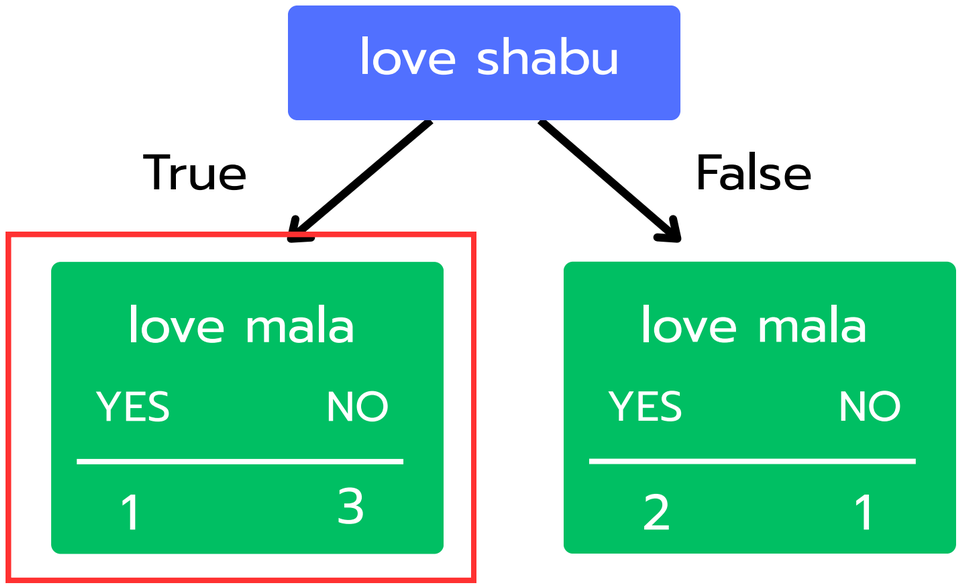
love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

Gini impurity

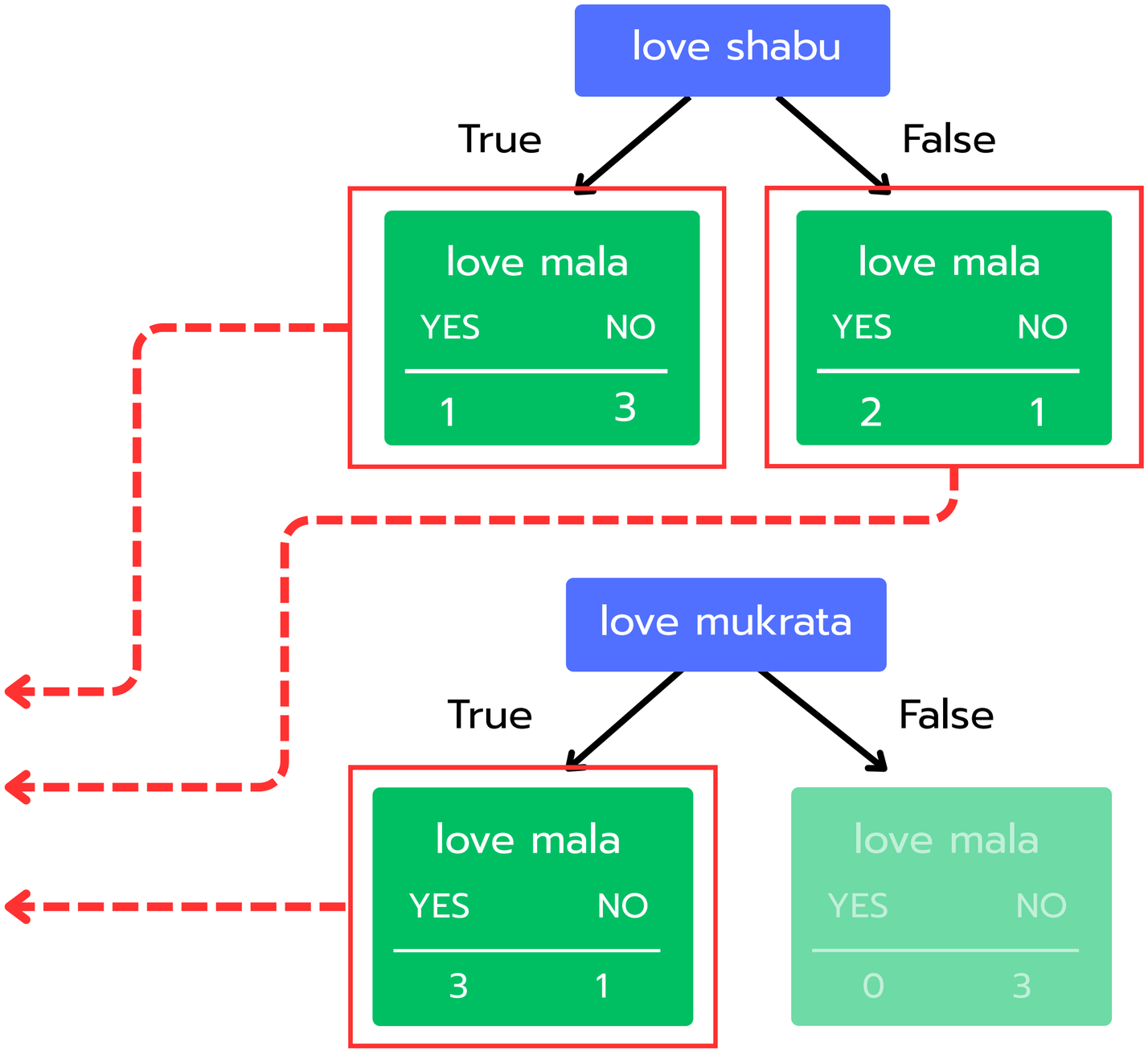
Gini Impurity = ค่าความปนเปของ class

ยิ่งน้อย → กลุ่มยิ่งบริสุทธิ์

love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

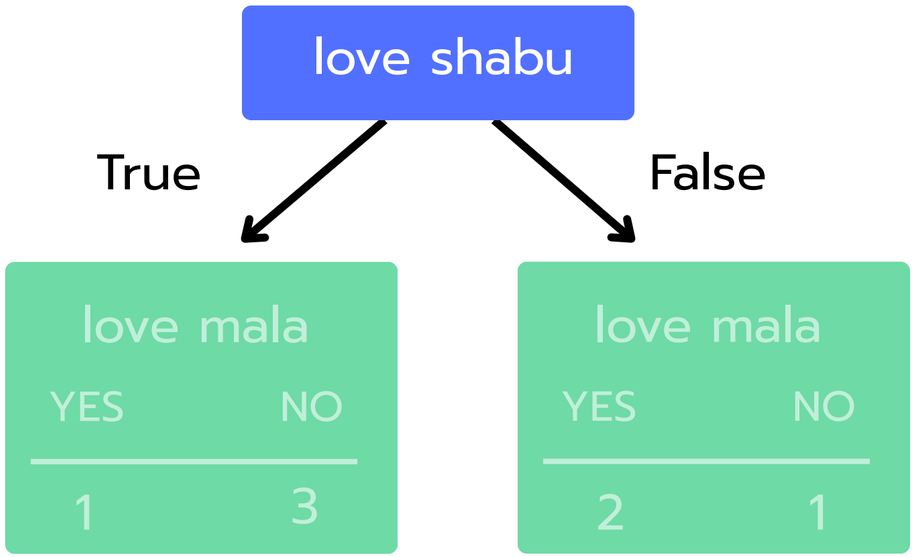


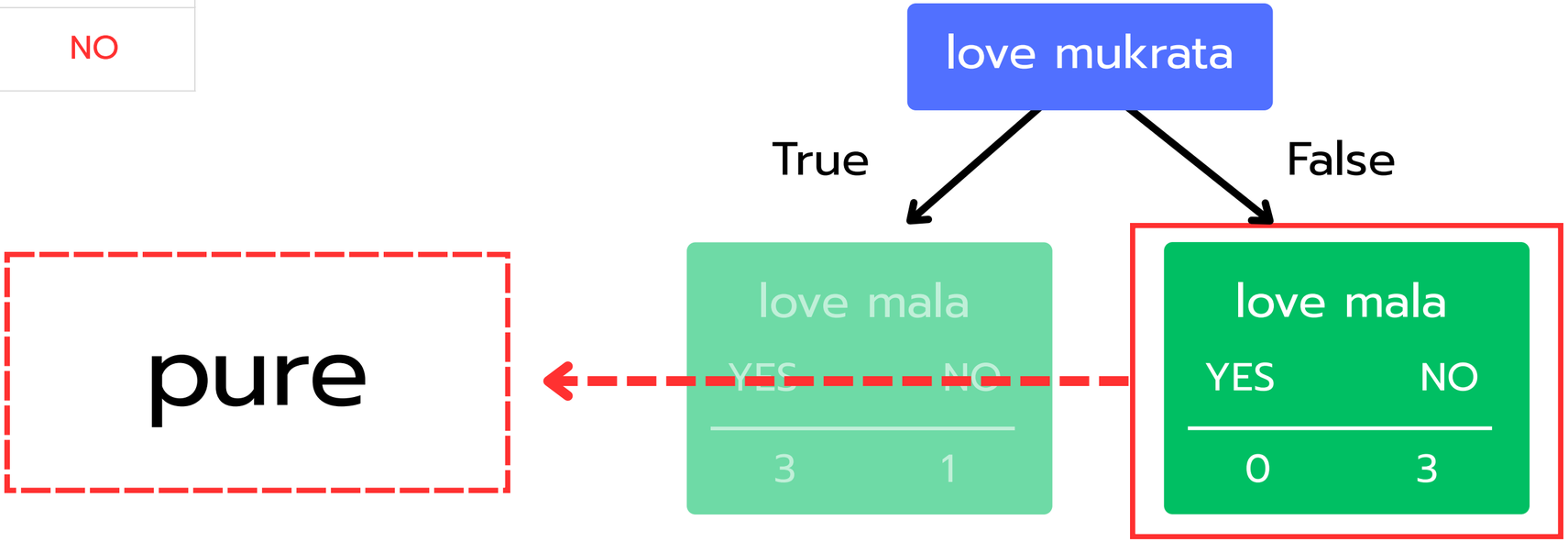
love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO



impure

love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO





love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

9.5

15

26.5

36.5

44

66.5

love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

9.5

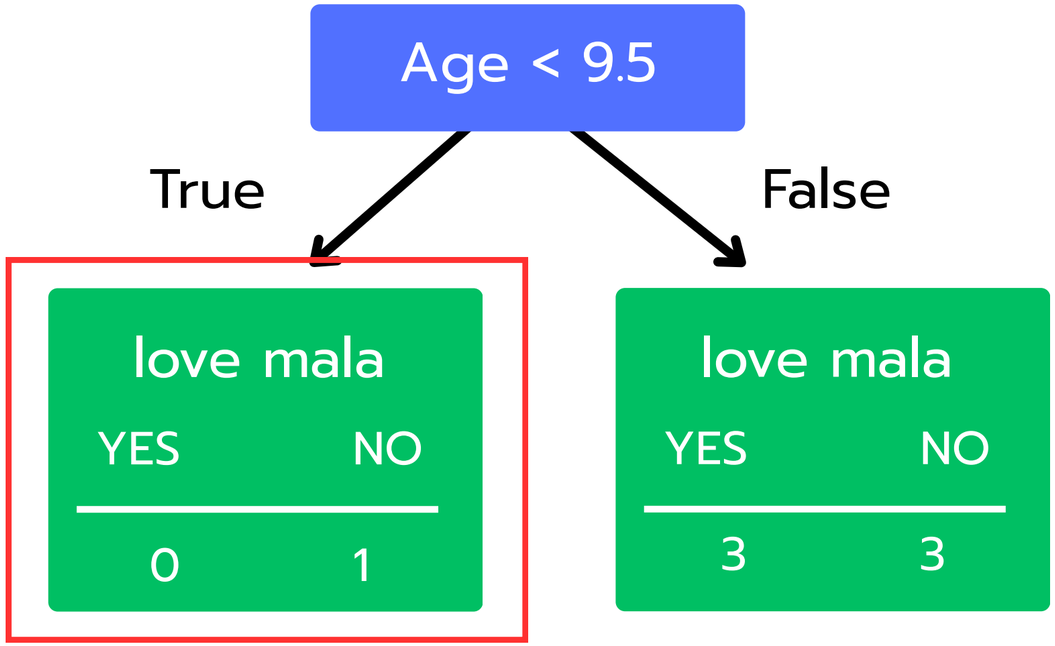
15

26.5

36.5

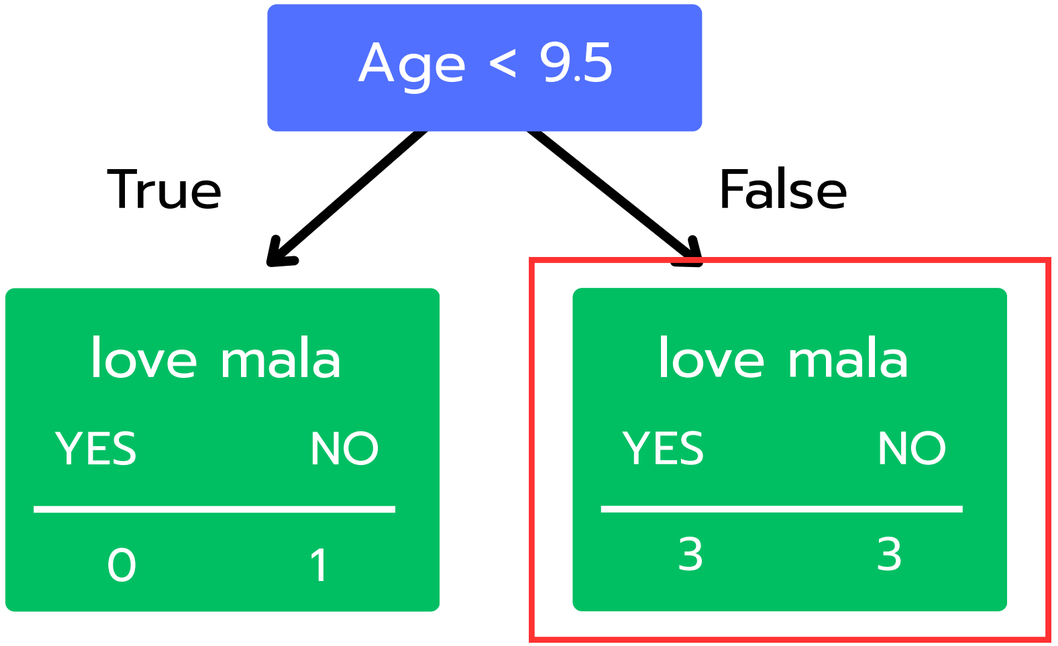
44

66.5



love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

9.5



love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

9.5

15

26.5

36.5

44

66.5

Age < 9.5

Age < 15

Age < 26.5

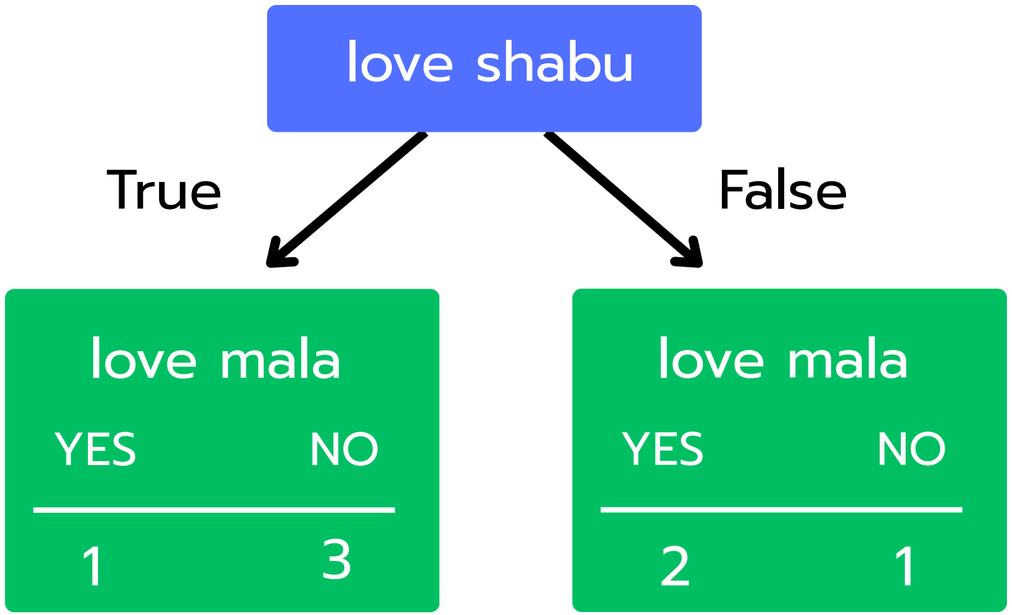
Age < 36.5

Age < 44

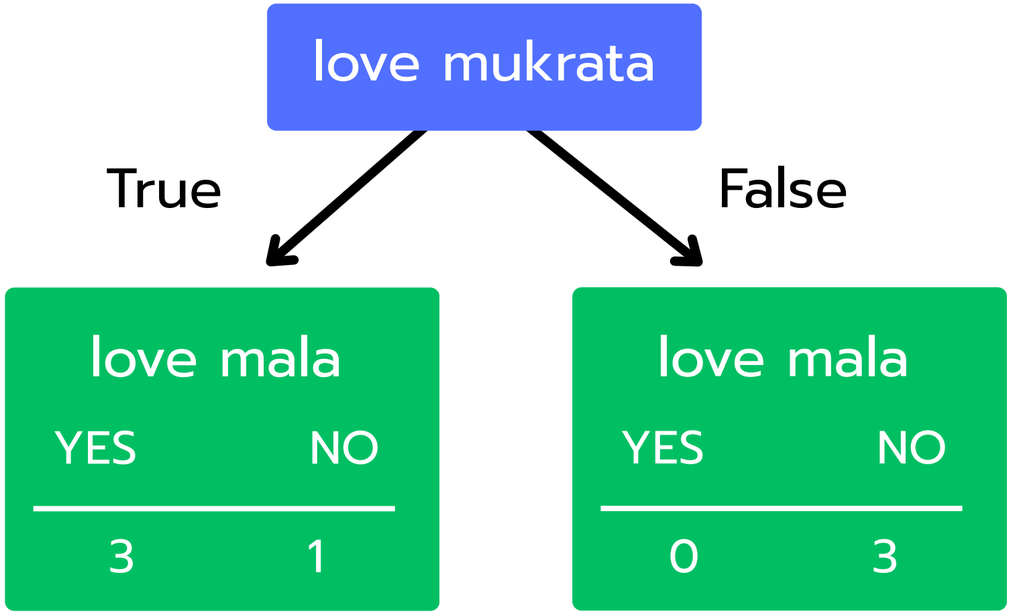
Age < 66.5

```
clf = DecisionTreeClassifier(criterion='gini')
```

Gini impurity = 0.405



Gini impurity = 0.214



love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

9.5 → Gini impurity = 0.429

15 → Gini impurity = 0.343

26.5 → Gini impurity = 0.476

36.5 → Gini impurity = 0.476

44 → Gini impurity = 0.343

66.5 → Gini impurity = 0.429

love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

9.5 → Gini impurity = 0.429

15 → Gini impurity = 0.343

26.5 → Gini impurity = 0.476

36.5 → Gini impurity = 0.476

44 → Gini impurity = 0.343

66.5 → Gini impurity = 0.429

love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

9.5 → Gini impurity = 0.429

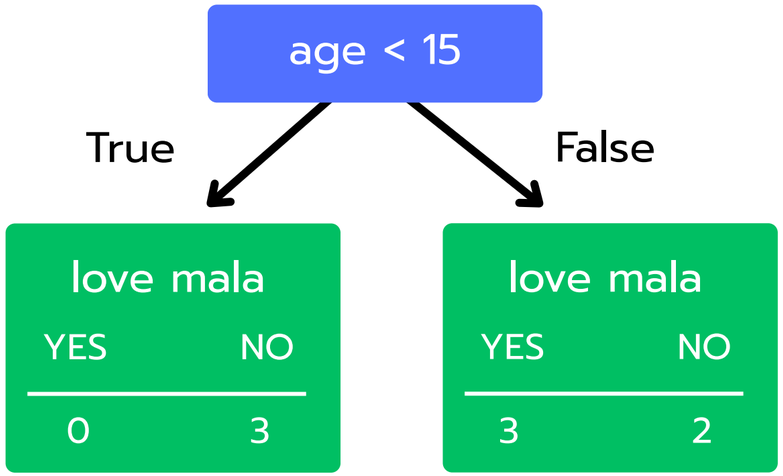
15 → Gini impurity = 0.343

26.5 → Gini impurity = 0.476

36.5 → Gini impurity = 0.476

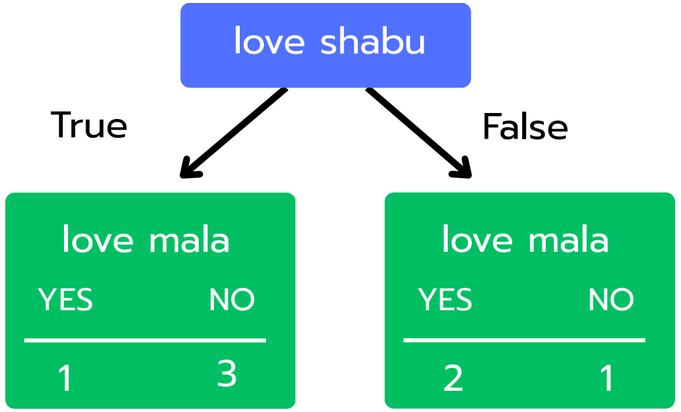
44 → Gini impurity = 0.343

66.5 → Gini impurity = 0.429

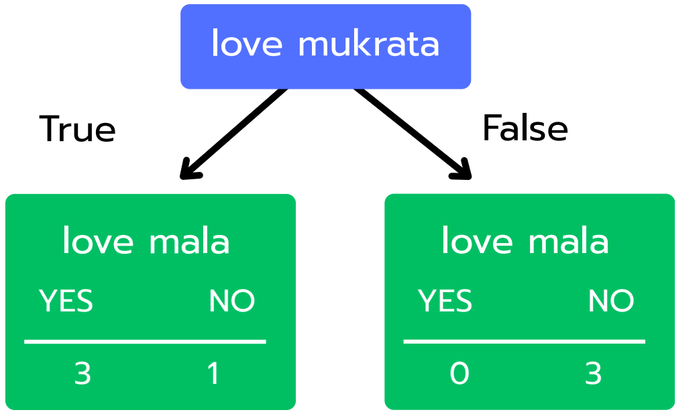


love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

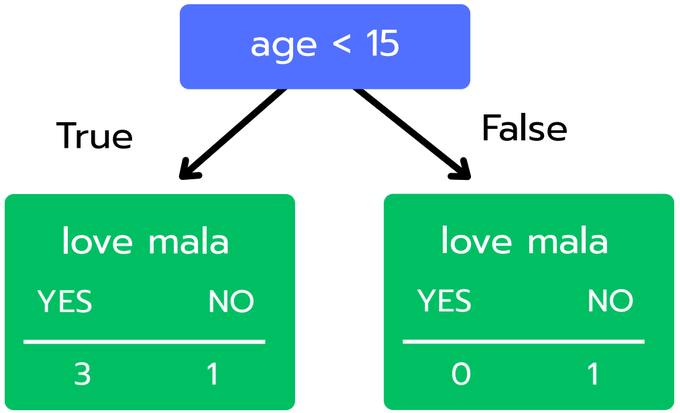
Gini impurity = 0.405

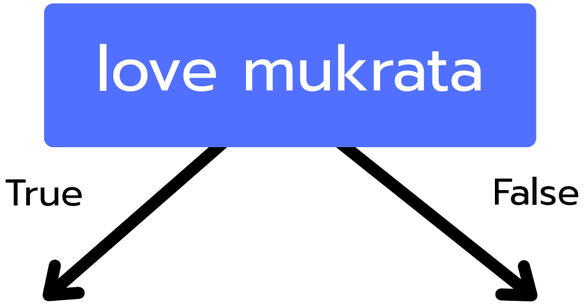


Gini impurity = 0.214

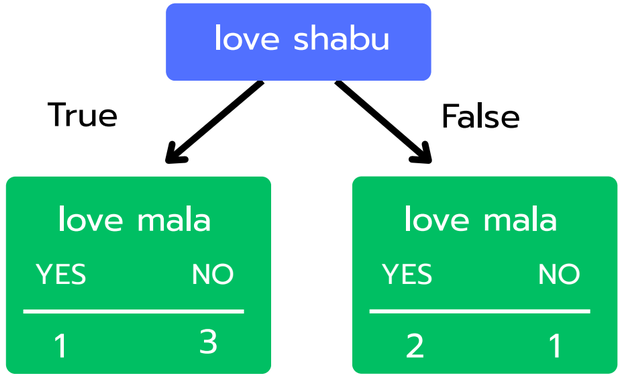


Gini impurity = 0.343

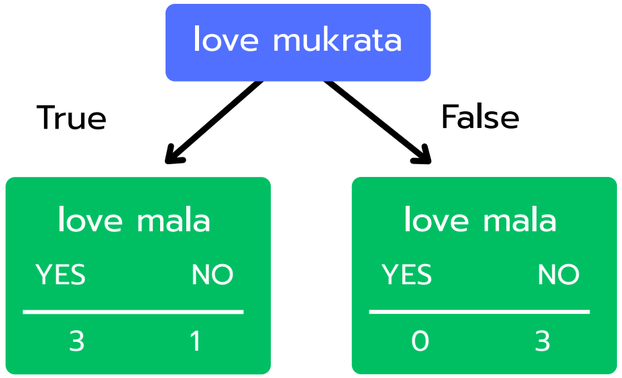




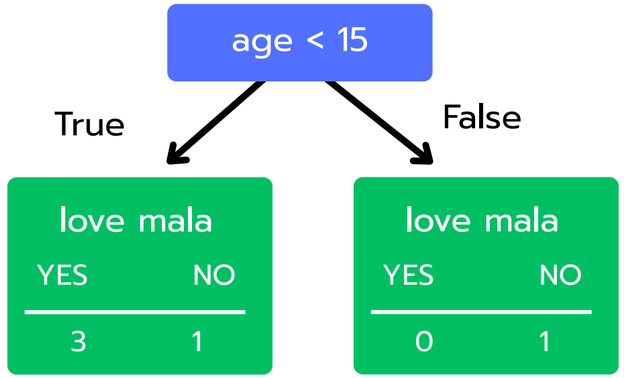
Gini impurity = 0.405



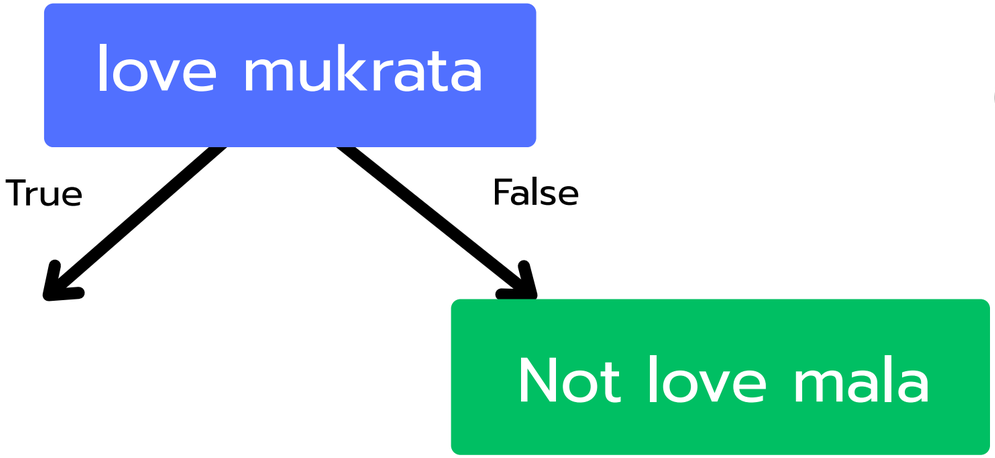
Gini impurity = 0.214



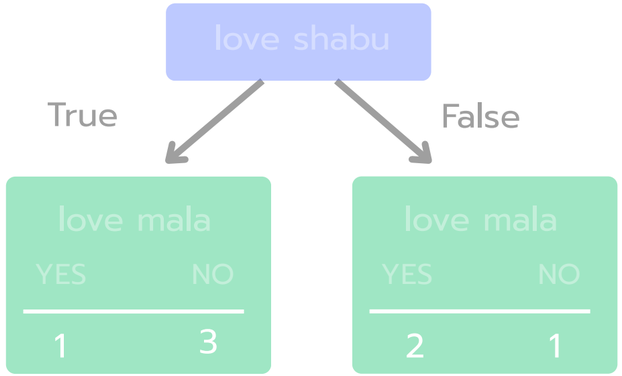
Gini impurity = 0.343



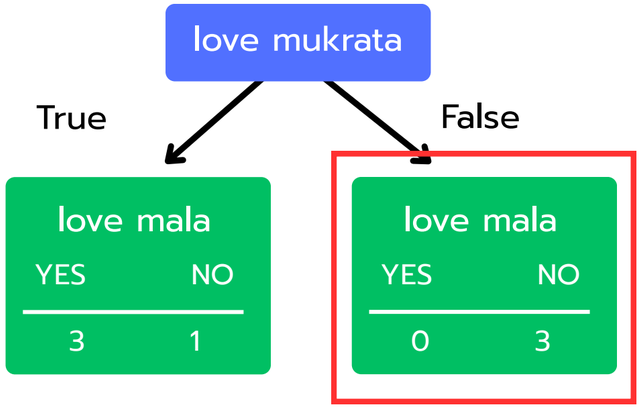
แบ่งข้อมูลตาม feature นั้น
และทำซ้ำขั้นตอน 1 - 4 จนกว่า ทุก node มี class เดียวกัน
หรือ ไม่มี feature เหลือ



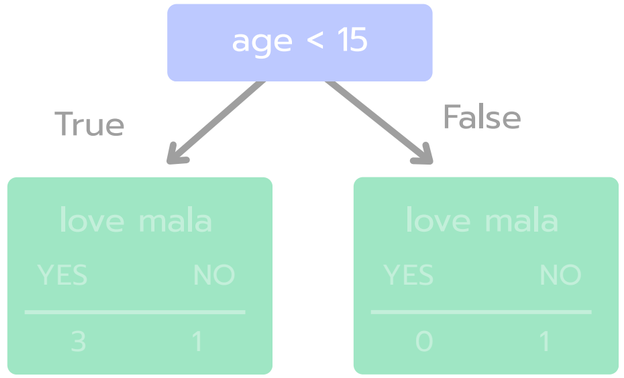
Gini impurity = 0.405

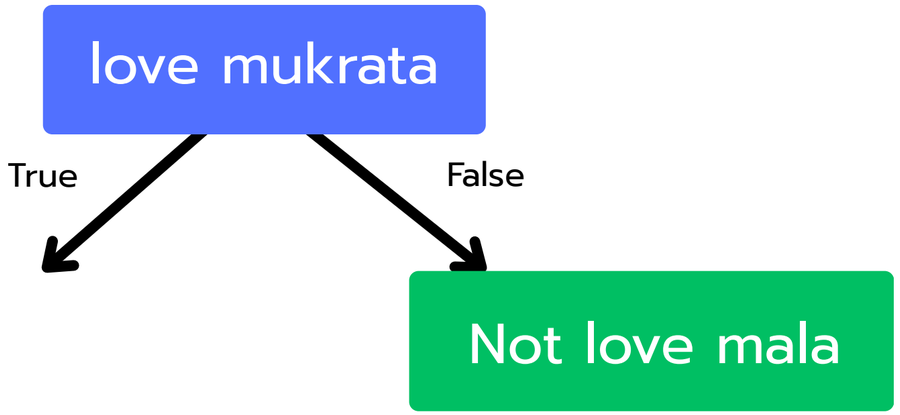


Gini impurity = 0.214



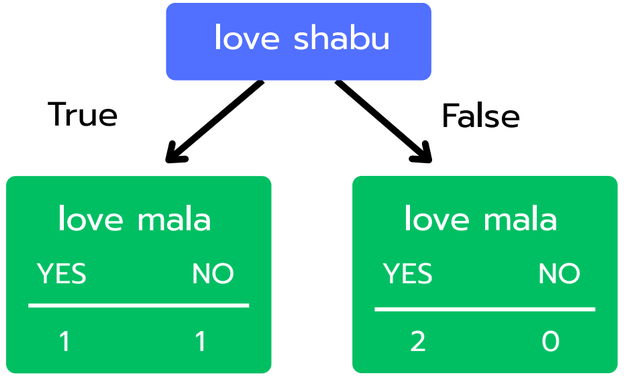
Gini impurity = 0.343

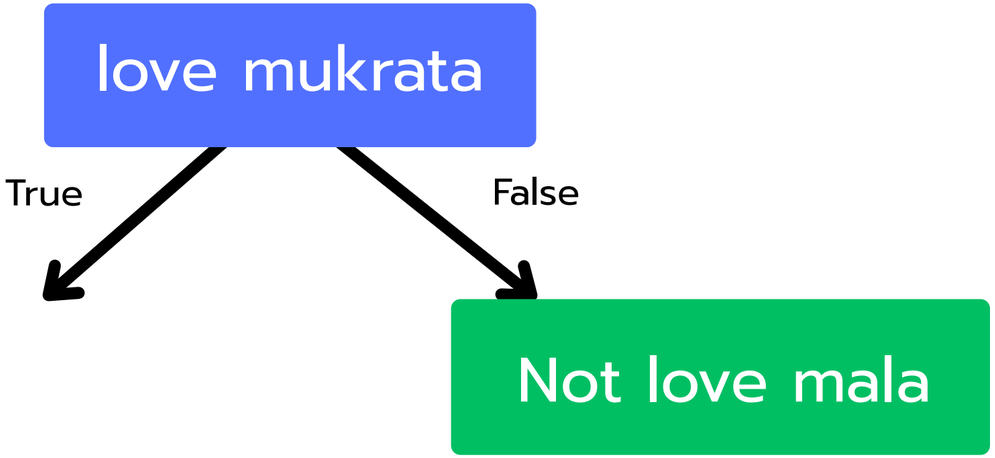




love shabu	love muukrata	age	love mala
YES	YES	7	NO
YES	NO	12	NO
NO	YES	18	YES
NO	YES	35	YES
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

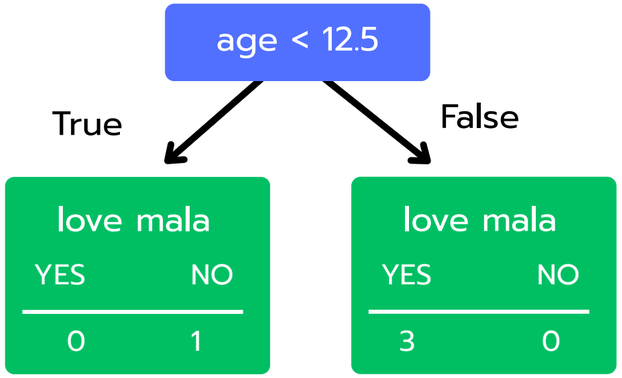
Gini impurity = 0.25

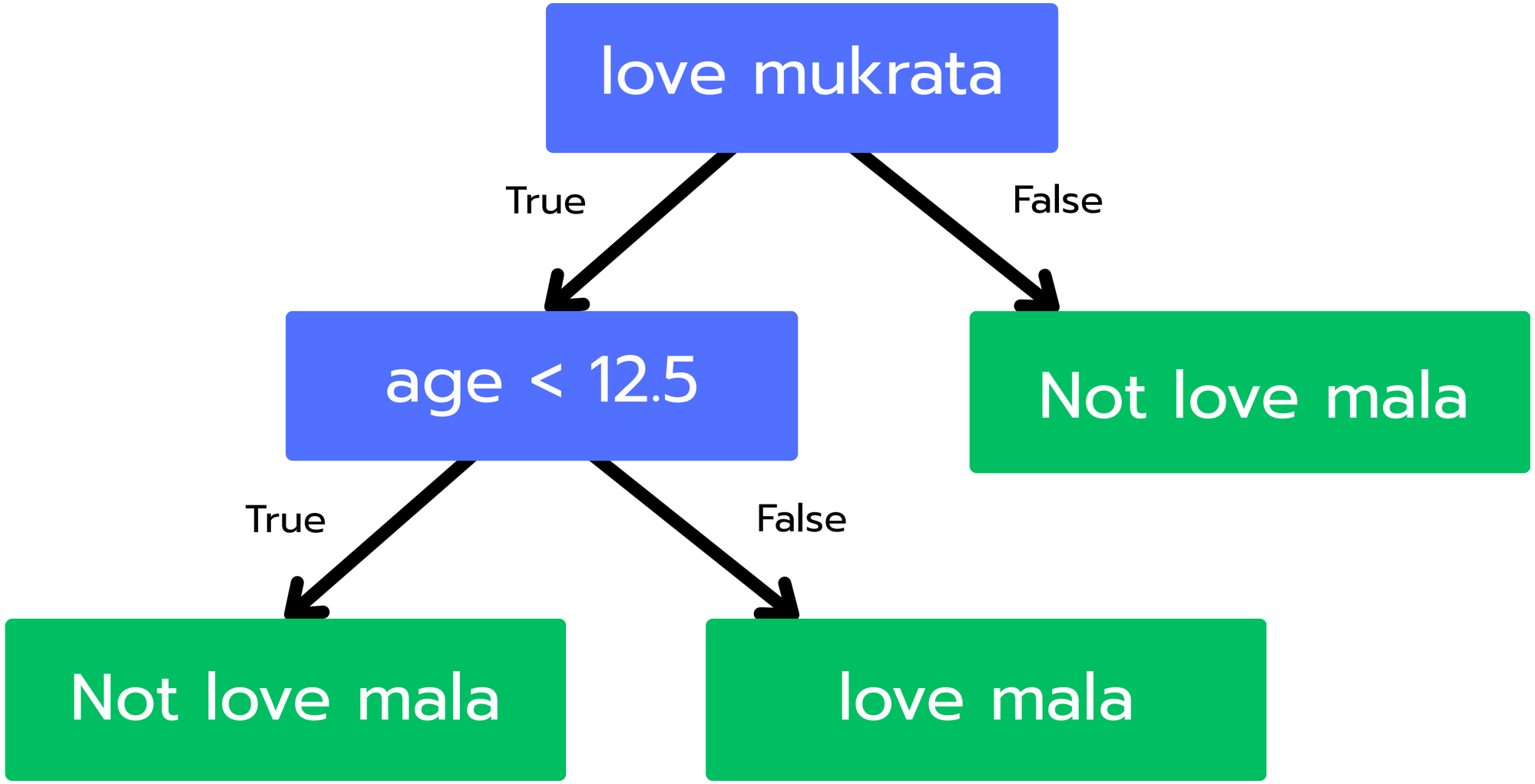




love shabu	love muukrata	age	love mala
YES	YES	7	NO
		12.5	
YES	NO	12	NO
NO	YES	18	YES
		26.5	
NO	YES	35	YES
		36.5	
YES	YES	38	YES
YES	NO	50	NO
NO	NO	83	NO

Gini impurity = 0




love mukrata
True
False
age < 12.5
Not love mala
True
False
Not love mala
love mala

Regression

ต่างจาก classification ยังไง?

นักเรียน	ชั่วโมงอ่านหนังสือ	ชั่วโมงนอน	คะแนน (0–100)	ผลสอบ (เกณฑ์ 50)
A	10	8	85	ผ่าน
B	2	5	40	ตก
C	8	7	75	ผ่าน
D	5	6	60	ผ่าน
E	1	4	30	ตก

Classification

สิ่งที่ทำนาย : น่าจะ “ผ่าน”

Regression

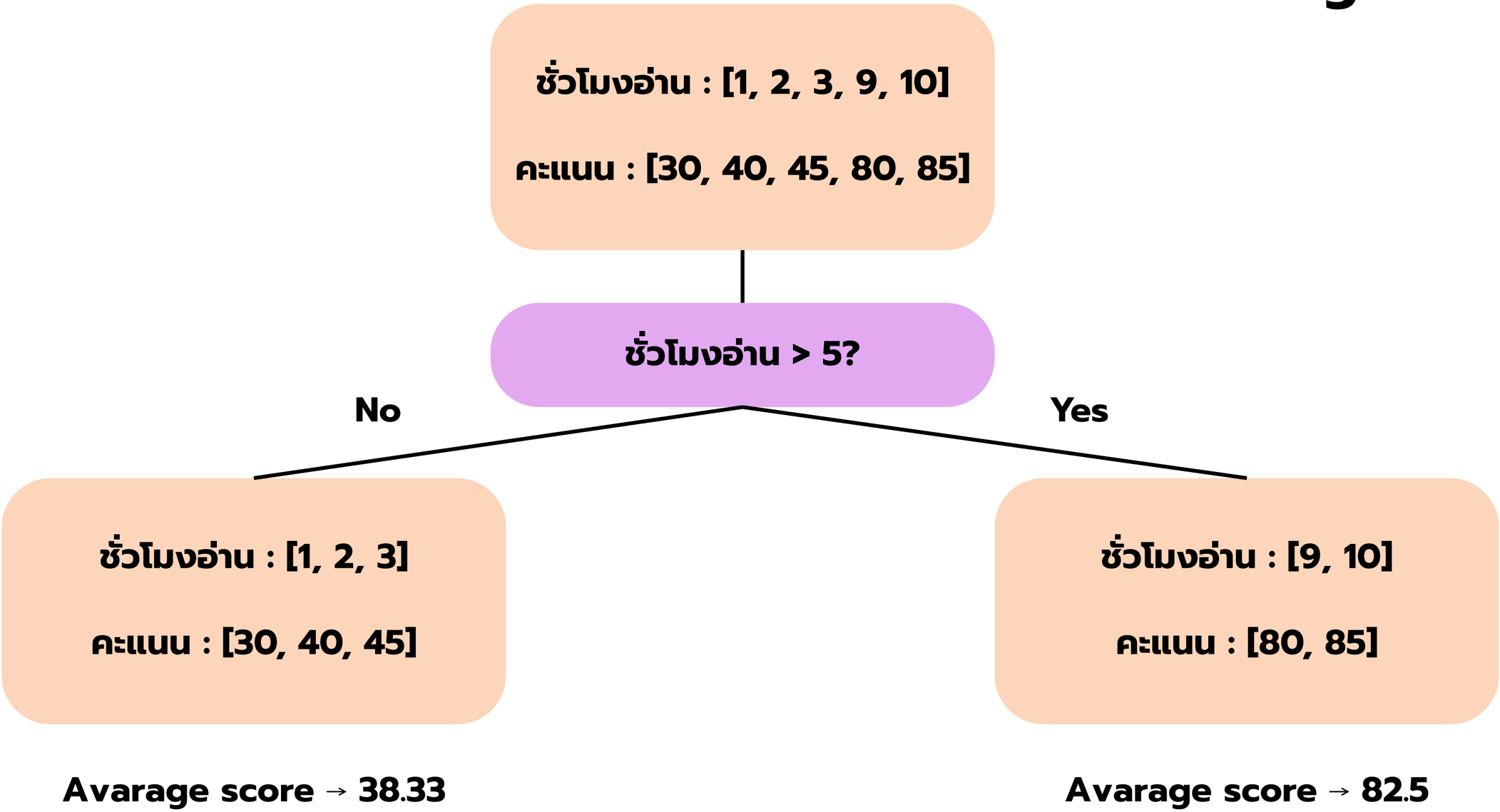
สิ่งที่ทำนาย : น่าจะได้ “73” คะแนน

Variance Reduction

Why?

นักเรียน	ชั่วโมงอ่านหนังสือ	ชั่วโมงนอน	คะแนน (0–100)	ผลสอบ (เกณฑ์ 50)
A	10	8	85	ผ่าน
B	2	5	40	ตก
C	8	7	75	ผ่าน
D	5	6	60	ผ่าน
E	1	4	30	ตก

Regression


ชั่วโมงอ่าน : [1, 2, 3, 9, 10]
คะแนน : [30, 40, 45, 80, 85]
ชั่วโมงอ่าน > 5?
No
Yes
ชั่วโมงอ่าน : [1, 2, 3]
คะแนน : [30, 40, 45]
ชั่วโมงอ่าน : [9, 10]
คะแนน : [80, 85]
Avarage score → 38.33
Avarage score → 82.5

ตรงไหน?

คะแนน : [30, 40, 45, 80, 85]

Regression

ชั่วโมงอ่าน > 5?

No

Yes

2, 3]

ชั่วโมงอ

Varience = 70

ถ้าเลือก 3 ลดเหลือ 50

ถ้าเลือก 7 ลดเหลือ 68

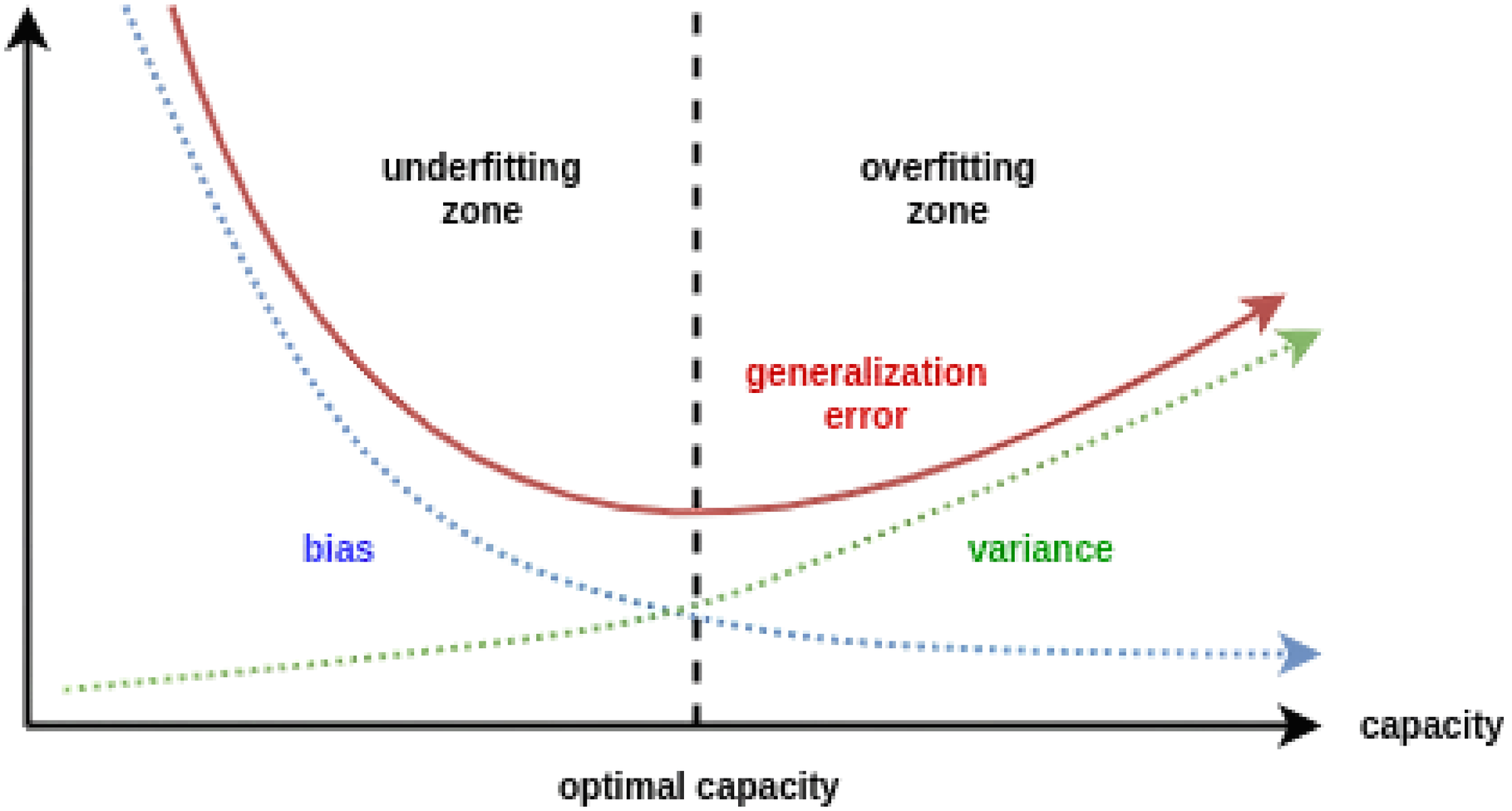
ถ้าเลือก 5 ลดเหลือ 40

Overfitting

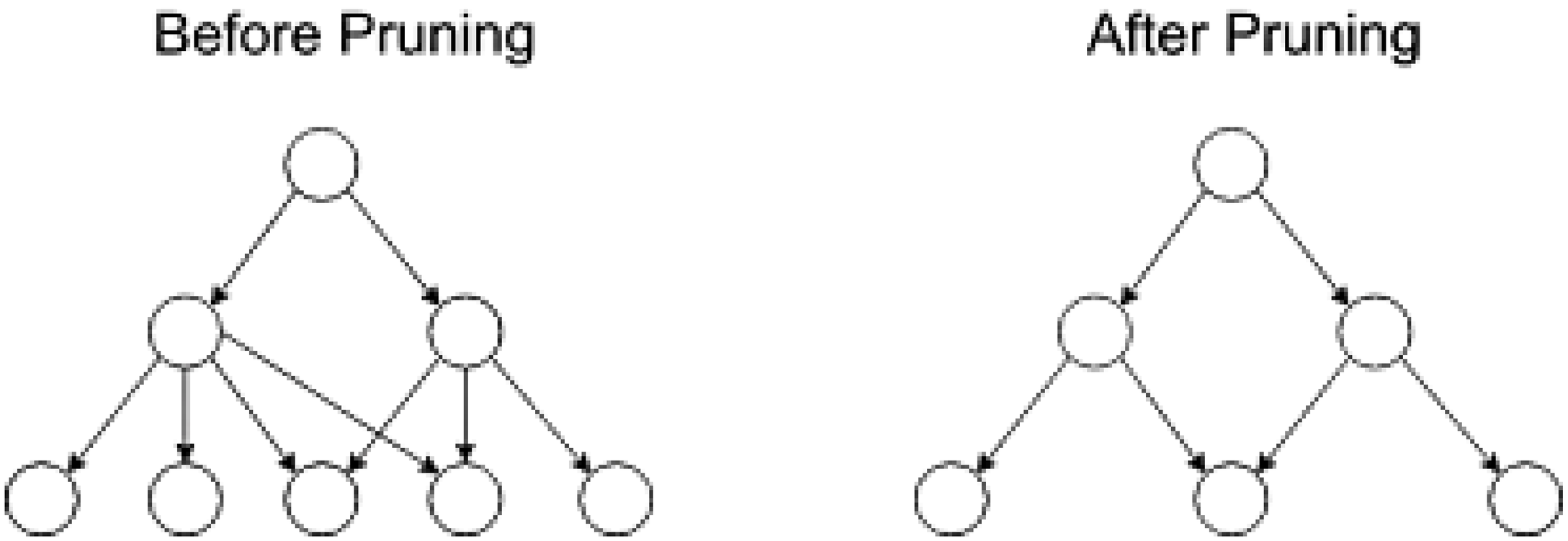
วิธีแก้

- Early stopping
- Post Pruning
- Random forest

- Early stopping

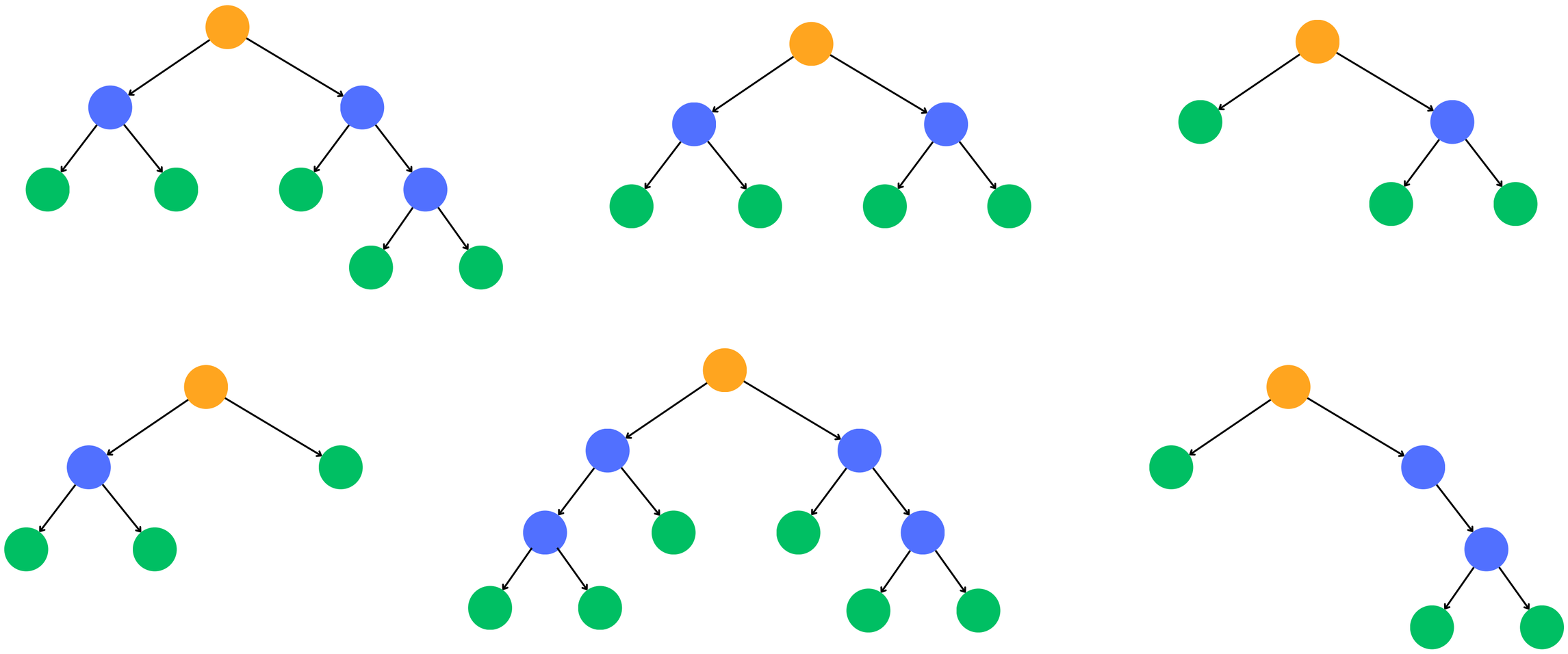


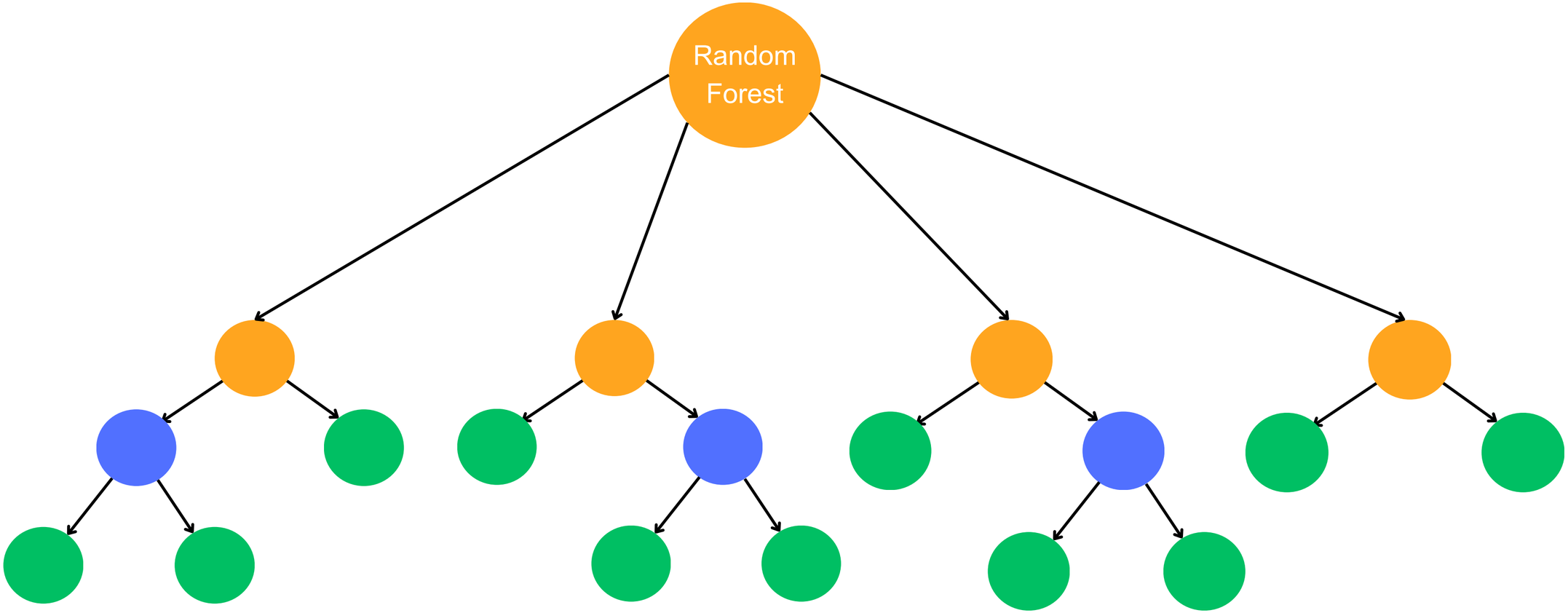
- Post Pruning



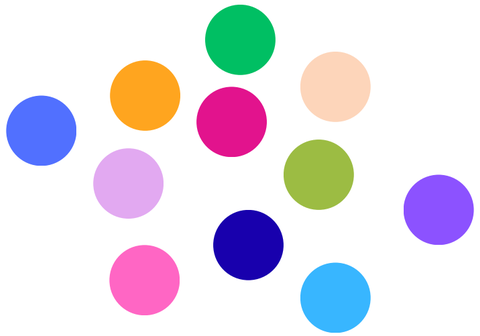
Random forest

A Machine Learning model of the Ensemble type that is built from multiple Decision Trees combined together.



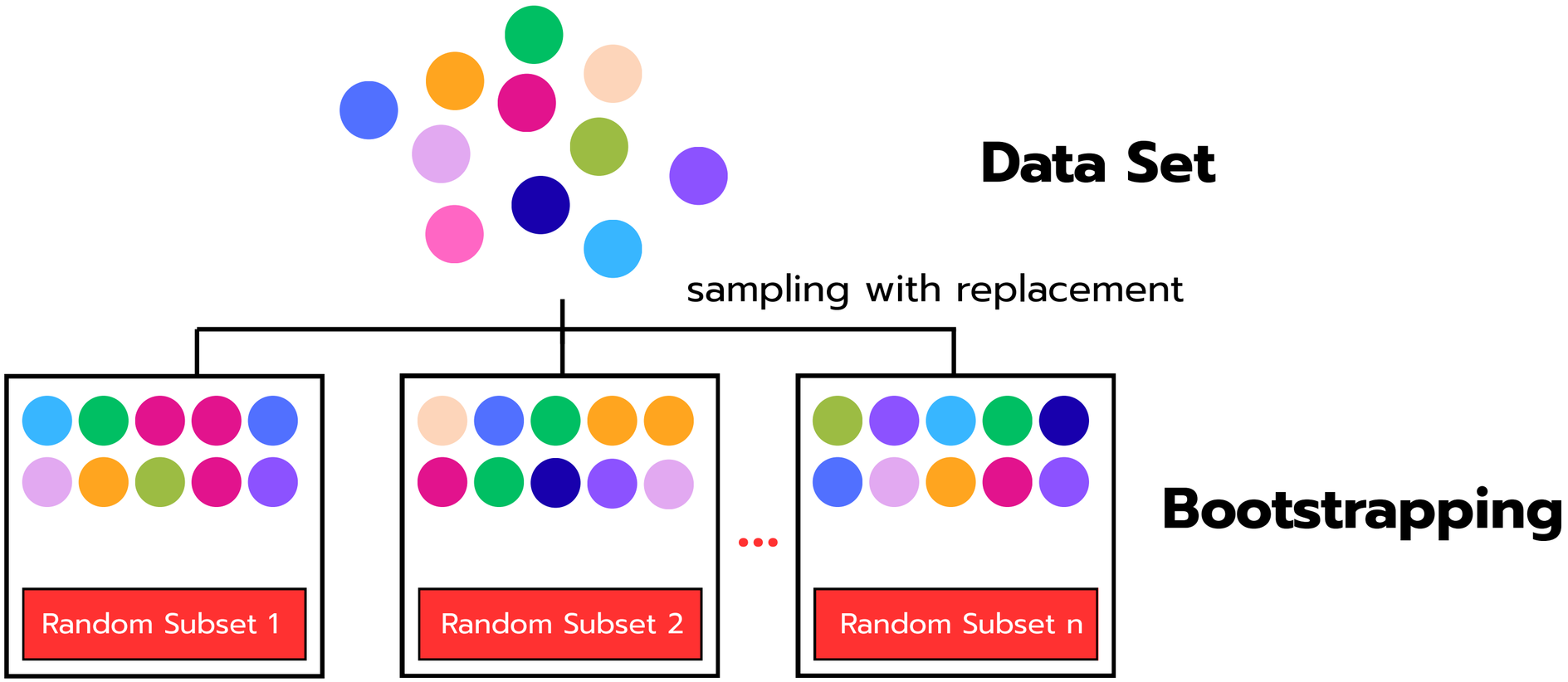

Random
Forest

Prepare Data



Original Data
(Data Set)

Bootstrapping

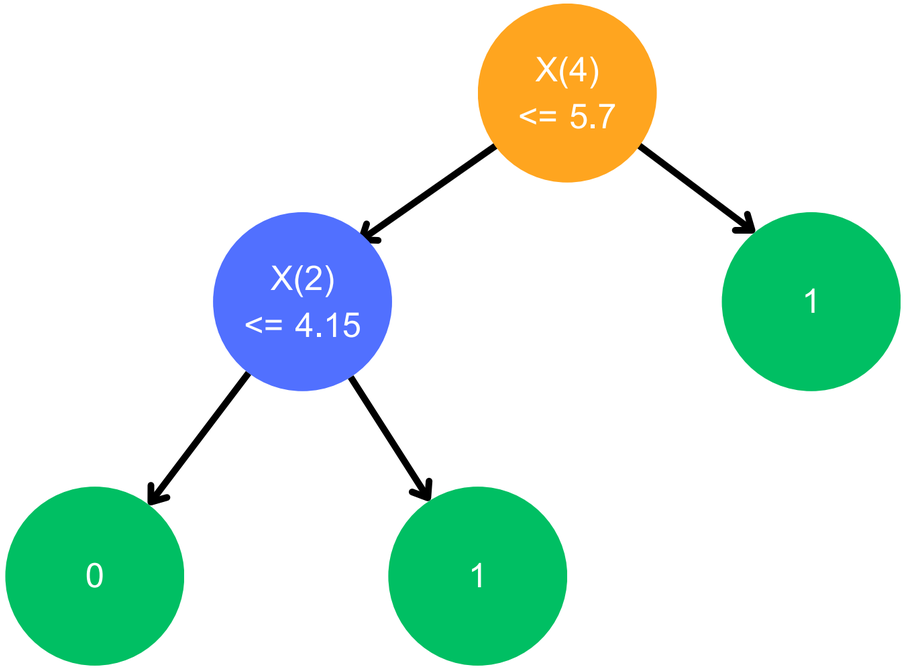

Data Set
sampling with replacement
Random Subset 1
Random Subset 2
Random Subset n
Bootstrapping

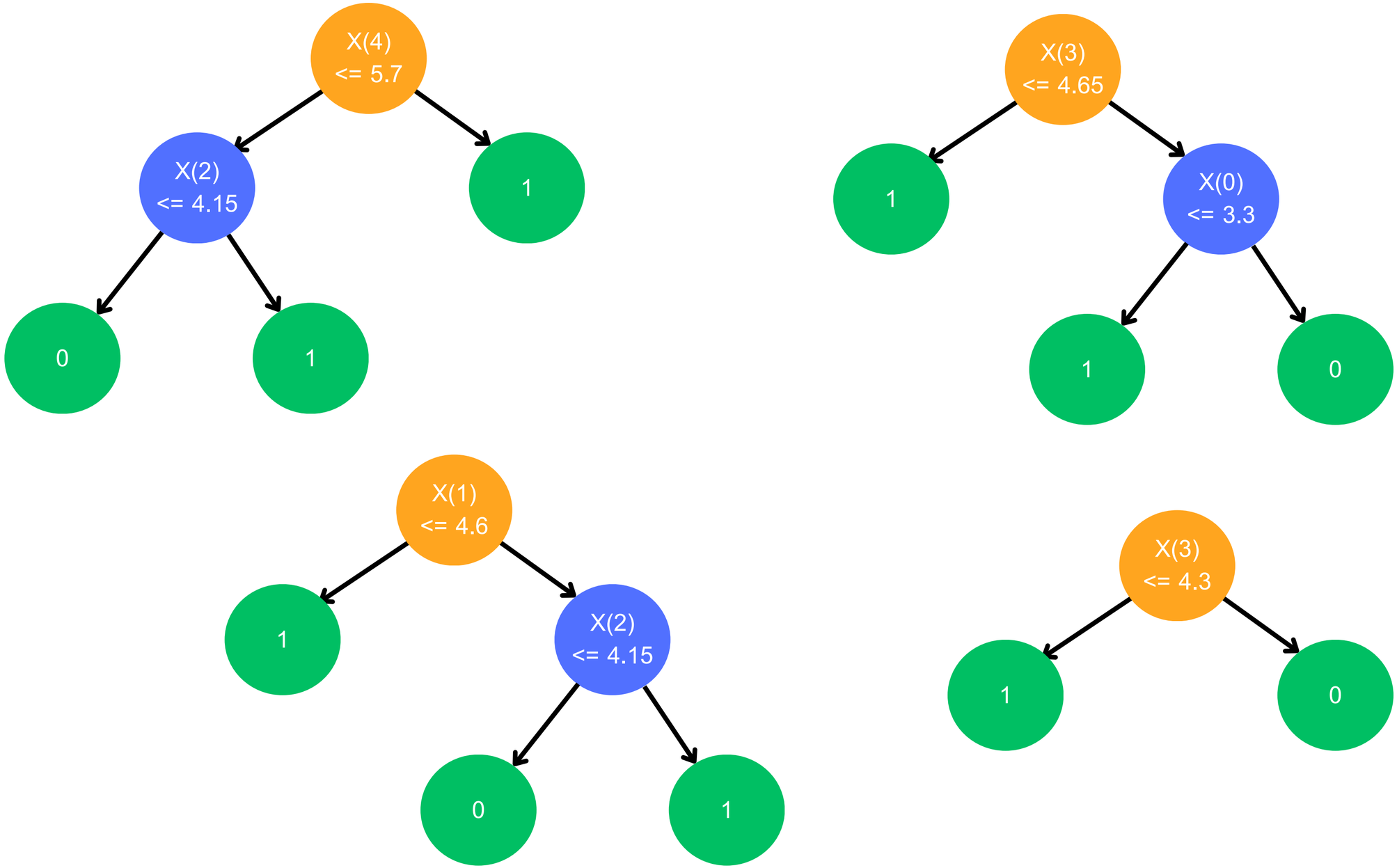
Building Decision Trees

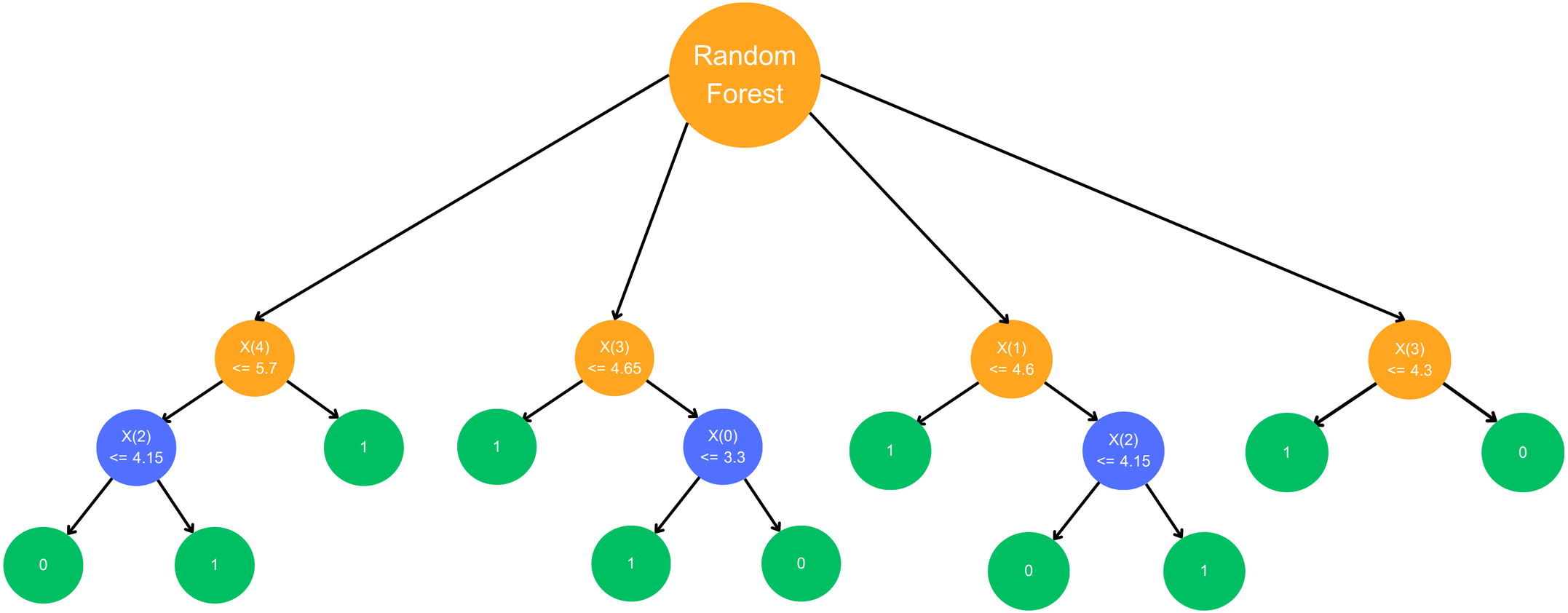
Zoom →

Feature | Target

ID	X(0)	X(1)	X(2)	X(3)	X(4)	y
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1
0	4.3	4.9	4.1	4.7	5.5	0
3	6.6	4.4	4.5	3.9	5.9	1
3	6.6	4.4	4.5	3.9	5.9	1
3	6.6	4.4	4.5	3.9	5.9	1



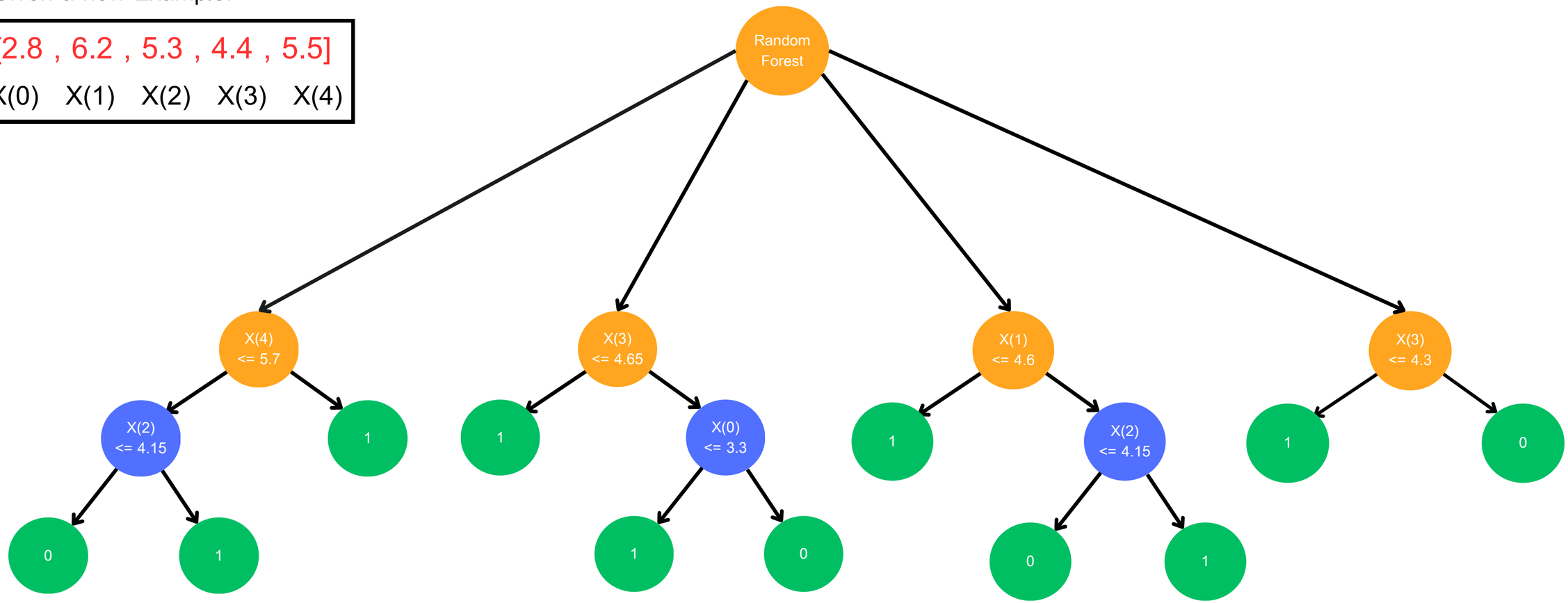

X(4)
<= 5.7
X(2)
<= 4.15
1
0
1
X(3)
<= 4.65
1
X(0)
<= 3.3
1
0
X(1)
<= 4.6
1
X(2)
<= 4.15
0
1
X(3)
<= 4.3
1
0


Random Forest
X(4) <= 5.7
X(3) <= 4.65
X(1) <= 4.6
X(3) <= 4.3
X(2) <= 4.15
1
1
X(0) <= 3.3
1
X(2) <= 4.15
1
0
0
1
1
0
0
1

Prediction

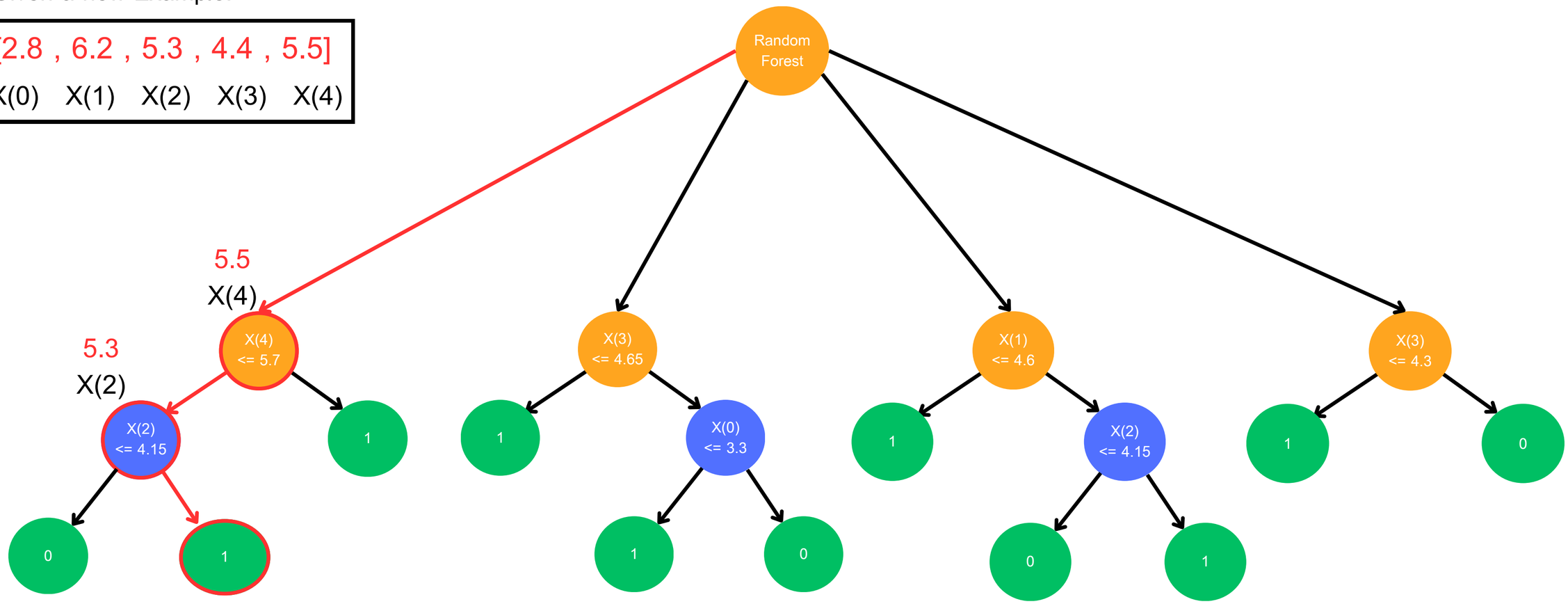
Given a new Example:

[2.8 ,	6.2 ,	5.3 ,	4.4 ,	5.5]
X(0)	X(1)	X(2)	X(3)	X(4)



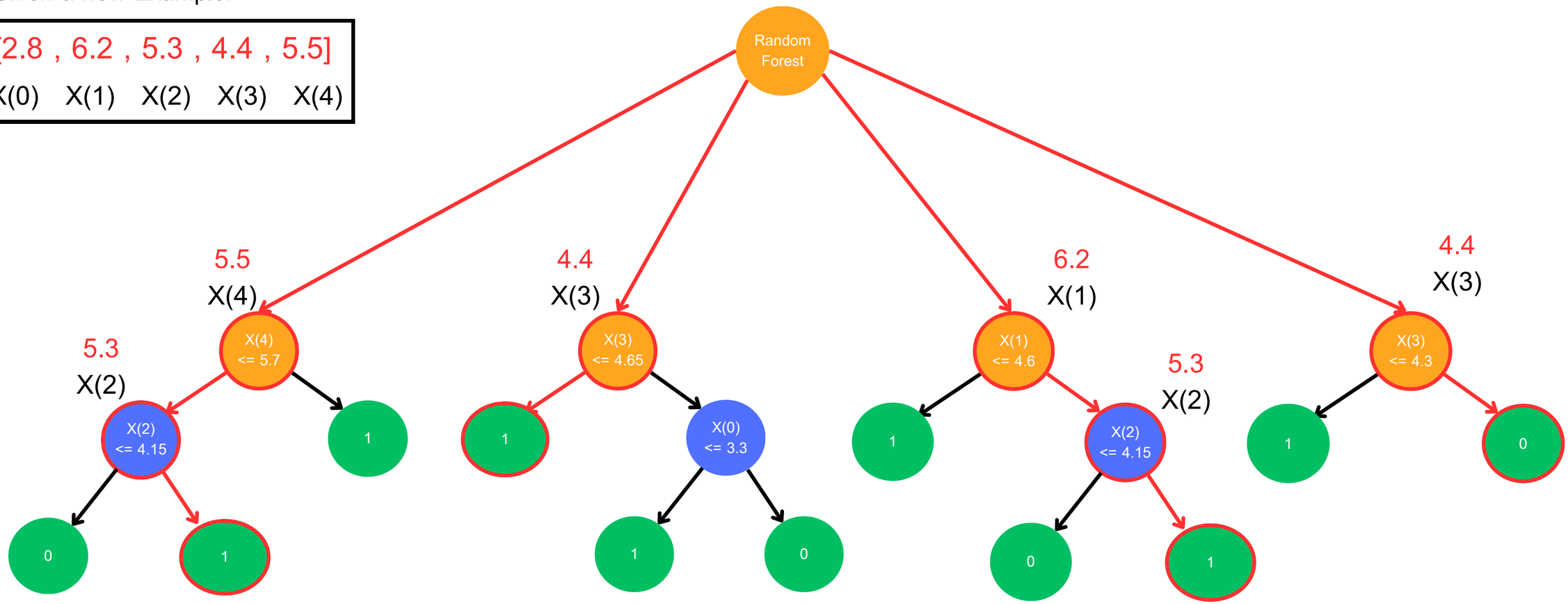
Given a new Example:

[2.8 ,	6.2 ,	5.3 ,	4.4 ,	5.5]
X(0)	X(1)	X(2)	X(3)	X(4)



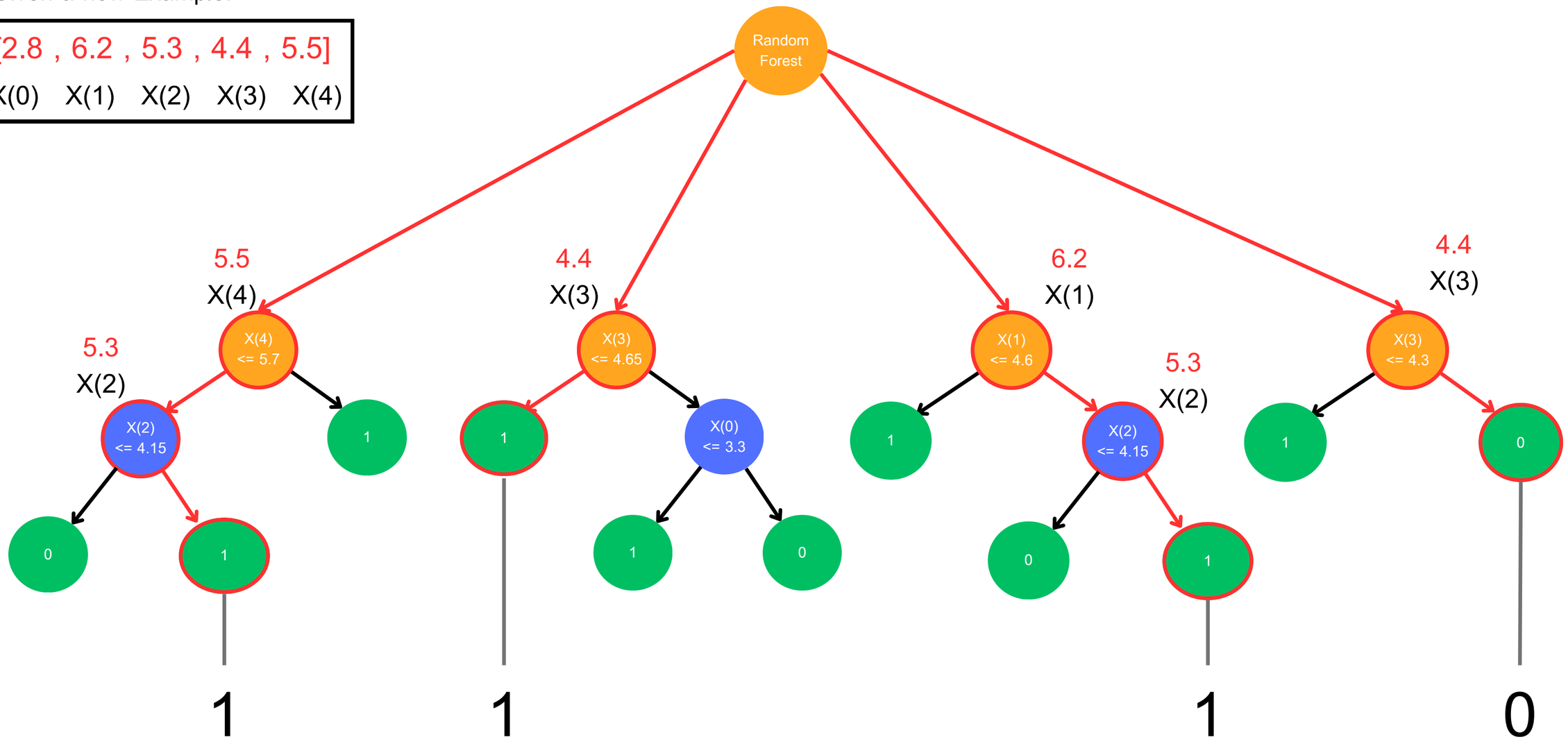
Given a new Example:

[2.8 ,	6.2 ,	5.3 ,	4.4 ,	5.5]
X(0)	X(1)	X(2)	X(3)	X(4)



Given a new Example:

[2.8 ,	6.2 ,	5.3 ,	4.4 ,	5.5]
X(0)	X(1)	X(2)	X(3)	X(4)

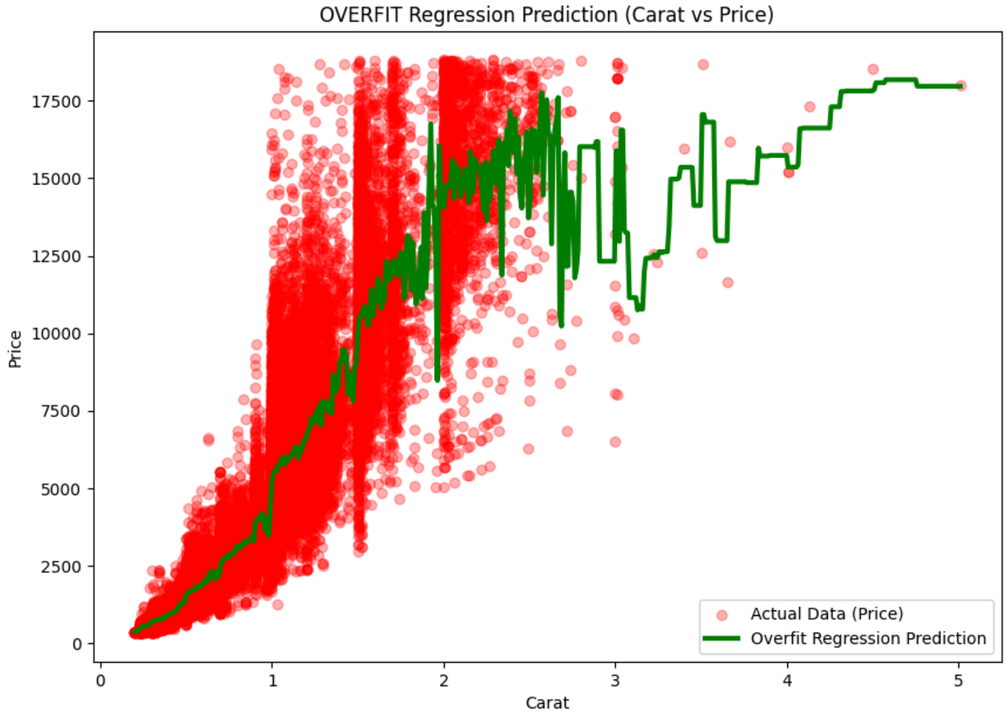


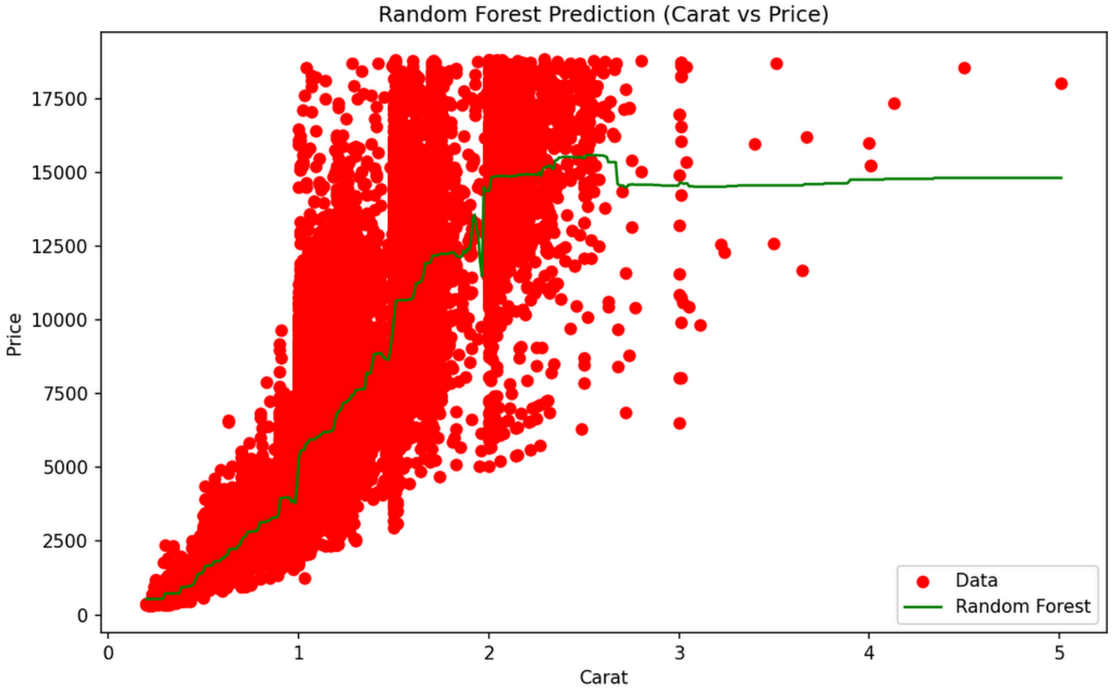
$$\boxed{1,\ 1,\ 1,\ 0}$$

$$\xrightarrow{\text{Majority Voting}}$$

$$1$$

DEMO


OVERFIT Regression Prediction (Carat vs Price)
Price
Carat
Actual Data (Price)
Overfit Regression Prediction


Random Forest Prediction (Carat vs Price)
Price
Carat
Data
Random Forest

THANK YOU